APOLLO 8	
FLIGHT PLAN VOL. I	
PART NO.	S / N
SKB 32100030-201	1006

FLIGHT PLAN

TIME	EVENT		REMARKS
-00:45	LMP:	FLT RCDR - RECORD	
-00:09	LCC:	IGNITION COMMAND	
-00:01		L/V ENGINE LTS (5) - OUT	
00:00	LCC:CDR:	<u>REPORT</u> LIFT-OFF	LIFT-OFF LT - ON, MET STARTS COUNT
		P11 AUTO	
00:02	CDR:	<u>REPORT</u> YAW MNVR	
00:11	CDR:	<u>REPORT</u> ROLL AND PITCH PROGRAM INITIATE	
00:28	CDR	<u>REPORT</u> ROLL COMPLETE	
00:42	MCC-H:CDR:	<u>REPORT</u> MARK MODE 1B	
00:50	LMP:	<u>REPORT</u> CABIN PRESS DECREASE	
01:17	CDR:	<u>REPORT</u> MAX Q	
01:50	MCC-H:CDR:	<u>REPORT</u> MARK MODE 1C	
02:00	MCC-H:CDR:	<u>REPORT</u> GO/NO GO FOR STAGING	
02:05	CDR:	<u>REPORT</u> INBOARD ENGINE CUTOFF	
02:31	CDR:	<u>REPORT</u> OUTBOARD ENGINE CUTOFF	LTS 1, 2, 3, & 4 - ON
02:32	CDR:	<u>REPORT</u> S-IC/S-II STAGING	LTS OFF

TIME	EVENT		REMARKS
03:00	CDR:	REPORT 2ND PLANE SEP	>65% THRUST-S-II SEP LIGHT OUT
03:07	CDR:	REPORT TWR JETT & MODE II	
03:25	CDR:	REPORT GUIDANCE INITIATE	
03:53	MCC-H:	REPORT TRAJECTORY AND GUID. GO/NO GO	
04:00	CMP:	REPORT S/C GO/NO GO	
05:00	LMP:	REPORT S/C GO/NO GO	
05:53	MCC-H:CDR:	REPORT S-IVB TO ORBIT CAPABILITY	
06:00	CDR:	REPORT S/C GO/NO GO	
06:15	LMP:	OMNI ANT-D	IF LAUNCH AZIMUTH <90°
07:00			
	CDR:	REPORT S/C GO/NO GO	
08:00			
08:20	MCC-H:CDR:	REPORT GO/NO GO FOR STAGING	
08:40	CDR:	REPORT S-II CUTOFF, S-II STAGING	

FLIGHT PLAN

TIME	EVENT		REMARKS
08:45	CDR:	REPORT S-IVB IGNITION	
09:00	CDR:	REPORT S/C GO/NO GO	
	MCC-H:	REPORT TRAJECTORY AND GUID. GO/NO GO	
09:50	MCC-H:CDR:	REPORT MARK MODE IV	
10:00	MCC-H:CDR:	REPORT GO/NO GO FOR ORBIT	
	MCC-H:	REPORT PREDICTED TIME OF SECO	
11:21	CDR:	REPORT SECO AND HP	
11:31	MCC-H:CDR:	REPORT ORBITAL GO/NO GO	
12:00	LMP:	FLT RCDR - OFF	

Time	Station	CDR	CMP	LMP	MCC-H
00:00	US	POST INSERTION CONFIG	POST INSERTION CONFIG SM RCS CK CM RCS CK C&W CK REMOVE HELMET & GLOVES	POST INSERTION CONFIG	
00:15	CYI	REMOVE HELMET & GLOVES ECS POST INSERTION CONFIG		REMOVE HELMET & GLOVES ECS POST INSERTION CONFIG	
	~~CYI~~	GDC ALIGN TO IMU	INGRESS LEB O_2 MAIN REG CK	EPS PERIODIC MONITOR ECS MONITOR CK SPS PERIODIC MONITOR	
00:30	TAN	MOUNT & INITIALIZE ORDEAL INSTALL COAS COAS HORIZON CK	JETTISON OPTICS COVER RECORD ΔAZ CORRECTION	~~PUGS TEST~~ ECS REDUNDANT COMP CK FC PURGE CK	VOICE UPDATE: ΔAZ CORRECTION
00:45	CRO		OPTICS CK IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID__ __,___ (cont'd)	BIOMED Sw - CENTER	
01:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	00:00 - 01:00	1/LPO	2-1

FLIGHT PLANNING BRANCH

MSC Form 1910 (OT) (Oct 68)

FLIGHT PLAN

Time	Station	CDR	CMP	LMP	MCC-H
01:00	HSK		STAR ANGLE DIFF __.__ TORQUE ANGLES: X __ __.__ __ __ Y __ __.__ __ __ Z __ __.__ __ __		
01:15					
01:30	US	SCS ATT REFERENCE COMPARISON CK	REPORT: GYRO TORQUE ANGLES RECORD ABORT BLOCK PAD (TLI +90 MIN AND TLI +4 HOUR)	BACKUP COMM CK	GIVE GO FOR COMM CK VOICE UPDATE: BLOCK DATA
01:45	CYI		RECORD TLI PAD	BIOMED Sw - RIGHT	VOICE UPDATE: TLI PAD P27 UPDATE: STATE VECTORS
02:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

2-2a

BURN STATUS REPORT

X X ☐ :	ΔTIG
X X :	BT
☐ .	V_{gx}
TRIM	
X X X	R
X X X	P
X X X	Y
☐	Vi
☐	$\dot{h}$
☐ .	h
☐ .	ΔV_c

REMARKS:

TLI
BURN CHART

	P or Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
TLI	10°/SEC SHUTDOWN	+45° SHUTDOWN	B/T+6 SEC & Vi=PAD VALUE	NO TRIM

TLI PREMATURE SHUTDOWN	
ha >60,000 nm	LUNAR ORBIT OR FLYBY (DEPENDING ON ΔV REQD
ha >22,000 nm	TWO PHASING MANEUVERS TO SEMI-SYNCHRONOUS ORBIT. DIRECT ENTRY
ha >4,000 nm	HIGH ALTITUDE ORBITS FOLLOWED BY DEBOOST TO 400 nm APOGEE
ha 100 - 4,000 nm	EITHER HI ALTITUDE (4,000 ha) OR LOW ALT. DEPENDING ON LANDMARKS

FLIGHT PLAN

GET	STATION	CDR	CMP	LMP	MCC-H
02:00			UNSTOW PHOTO EQUIP B3 16mm DAC 18mm LENS RT ANG MIRROR 16mm C-EX MAG PWR CABLE 70mm CAM 80mm LENS 70mm C-MAG		
	TAN				
02:15		TLI PREP EMS ΔV TEST	TLI PREP TRANS TO COUCH	TLI PREP S-BAND XPONDER - SEC	
	CRO	GO/NO-GO FOR PYRO ARM GO/NO-GO FOR TLI			GO/NO-GO
02:30		GDC ALIGN AND DRIFT CK		BIOMED Sw - LEFT	
		TB-6			
02:45	REDHAW	P47 BURN ATT CK $\boxed{\text{TLI}}$	$\boxed{\text{GETI = 2:50:31}}$	FLT RCDR - RECORD	
		SECO S-IVB INERTIAL SECO +20 SEC S-IVB TO LH,ORB RATE,HEADS DOWN		FLT RCDR - OFF INITIATE BATT B CHARGING (FOLLOWED BY BATT A)	
03:00					

GET	CDR	CMP	LMP	MCC-H
03:00	TLI BURN STATUS REPORT	RECORD GET SEP MNVR INIT		VOICE UPDATE: GET OF SEP MNVR INIT
	S-IVB MNVR TO SEP ATT	TRANS CSM STATE VECTOR TO LM SLOT		
03:15	GO/NO-GO FOR 90-MIN ABORT			GO/NO-GO
	TRANSPOSITION FROM S-IVB +X FOR 1 fps, COAST FOR 1 MIN, -X FOR 0.5 fps, PITCH UP 4°/SEC		FLT RCDR - RECORD	
		R13 SPOTMETER	FLT RCDR - OFF NONESS BUS - OFF	
03:30	FLY FORMATION	DOFF & STOW PGA	PHOTOGRAPH S-IVB 16/18/C-EX, 1/250, f11, 6 fps (1 MAG) 2/80/C, 1/250, SPOT (10 EXP)	
	MNVR TO LOCAL VERTICAL -X RADIALLY UPWARD 1.5 fps			
03:45		REINITIALIZE W MATRIX R1 +00094 R2 +00057 R3 +.00003	TERMINATE WATER BOILER EVAP	
		TRANS CSM STATE VECTOR TO LM SLOT		
04:00				

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	03:00 - 04:00	1/TLC	2-4

MSC Form 2910 (OT) (Oct 68) FLIGHT PLANNING BRANCH

FLIGHT PLAN

GET	CDR	CMP	LMP	MCC-H
04:00		IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID __ __ STAR ANGLE DIFF __ __ TORQUE ANGLES: X __ __ . __ __ __ Y __ __ . __ __ __ Z __ __ . __ __ __	REPORT PERSONAL RADIATION DOSIMETER READINGS	
04:15	GDC ALIGN TO IMU	CISLUNAR NAVIGATION P23 TRN BIAS 1. STAR 14 ENH 00110 STAR __ __ E __ H 1 SET - 3 MARKS EACH	Record Trunion Bias, ΔR, ΔV, +TA for all sightings.	
04:30		2. STAR 15 EFH 00120 STAR __ __ E __ H 2 SETS - 3 MARKS EACH		
04:45		3. STAR 16 EFH 00120 STAR __ __ E __ H 2 SETS - 3 MARKS EACH		
05:00				

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

Time		CDR	CMP	LMP	MCC-H
05:00	MSFN	DOFF & STOW PGA	GROUND TRACK DET P21	RECORD BLOCK DATA (TLI +4 HRS & TLI +11 HRS)	VOICE UPDATE: BLOCK DATA
05:15		MNVR TO PTC ATT P ~~331~~ 56 Y ~~331~~ 340 P Y ~~roll~~ 0.1~~°/SEC~~		BIOMED Sw - CENTER	
05:30				DOFF & STOW PGA	
05:45			KEY V64	S-BAND HIGH GAIN ANT CHECKOUT	COMM TEST MODES: 4.2(R,V.R,V, HBR) 6.2(R,V,V.R,V, HBR) .2 FM DSE DUMP
06:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	05:00 - 06:00	1/TLC	2-6

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

Time		CDR	CMP	LMP	MCC-H
06:00	MSFN				
06:15					
06:30		EAT PERIOD	EAT PERIOD	EAT PERIOD	
06:45					
07:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	06:00 - 07:00	1/TLC	2-7

CDR | CMP | LMP | MCC-H

07:00

(R, V. R, V, LBR)
(R, V, U. R, V, LBR)
(R, U. R, V, HBR)
(UVB. R, V, HBR)

PERFORM
COMM TESTS
MODES: 4.3
6.3
5.2
8.2

07:15

M
S
F
N

07:30

LiOH CANISTER CHANGE
(CARTRIDGE NO. 3 FROM
~~A3~~ B5 INTO CANISTER A)

OPEN COOLANT CONTROL
ATTENUATION PANEL
EVAP WATER CONT SEC
VLV - OFF

07:45

CLOSE COOLANT CONTROL
ATTENUATION PANEL

WASTE STOWAGE VENT -
CLOSED
BAT VENT VLV - OPEN
(UNTIL SERVICE
METER = 0)

P27 UPDATE:
STATE
VECTOR
TGT LOAD

08:00

BAT VENT VLV - CLOSED

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	07:00 - 08:00	1/TLC	2-8

BURN STATUS REPORT

X X ☐ : ΔTIG

X X : BT

☐ . V_{gx}

TRIM

X X X R

X X X P

X X X Y

☐ . V_{gx}

☐ . V_{gy}

☐ . V_{gz}

☐ . ΔV_c

X X X FUEL

X X X OX

X X X UNBALANCE

REMARKS:

MCC'S

BURN CHART

	P or Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
MCC(ALL)	10°/SEC TAKEOVER	10° TAKEOVER	B/T +1 SEC	TRIM TO 0.2 fps

Time	CDR	CMP	LMP	MCC-H
08:00	MNVR TO P52 ATT	IMU REALIGN P52 OPTION 3 - REFSMMAT AND DRIFT CK STAR ID ___ ___ STAR ANGLE DIFF ___.___ TORQUE ANGLES X ___ ___.___ ___ ___ Y ___ ___.___ ___ ___ Z ___ ___.___ ___ ___	RECORD MNVR PAD + TLI + 11 HR PAD	VOICE UPDATE: MNVR PAD + TLI + 11 HR PAD
08:15	V47 TRANS LM STATE VECTOR TO CSM SLOT →		BIOMED Sw - RIGHT	
08:30 (MSFN)	EXT ΔV P30 SPS/RCS THRUST P40/P41 MNVR TO BURN ATT	SXT STAR CK TRANS TO COUCH		PIPA BIAS CK
08:45	EMS ΔV TEST GDC ALIGN To IMU	SM RCS MON CK		
TLI +6 HRS 09:00	MCC$_1$ ΔV=NOMINALLY ZERO			

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	08:00 - 09:00	1/TLC	2-9

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

Time	CDR	CMP	LMP	MCC-H
09:00	V66 TRANS CSM STATE VECTOR TO LM SLOT → MCC$_1$ BURN STATUS REPORT MNVR TO SIGHTING ATT	SM RCS MON CK TRN BIAS **CISLUNAR NAVIGATION P23** 1. STAR 15 ELDMK 10 LAT 28.876°N LONG/2 56.292°W ALT 000.01 STAR ___ ___ ELDMK ___ ___ LAT ___ ___.___ ___ ___ LONG/2 ___ ___.___ ___ ___ ALT ___ ___.___ ___ ___ 2 SETS	SPS MONITOR CK ~~INITIATE BAT CHARGE~~	
09:30 (MSFN)	MNVR TO PTC ATT P ~~331~~ 56 Y ~~331~~ 340 P ___ Y ___ ROLL 0.1°/SEC	2. STAR 16 ELDMK 10 LAT 28.876°N LONG/2 56.292°W ALT 000.01 STAR ___ ___ ELDMK ___ ___ LAT ___ ___.___ ___ ___ LONG/2 ___ ___.___ ___ ___ ALT ___ ___.___ ___ ___ GROUND TRACK DET P21		
10:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

10:00

10:30 — CDR: EAT PERIOD

11:00 — MSFN

11:30 — CDR: SLEEP PERIOD (7 HOURS) — LMP: BIOMED Sw - CENTER

12:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	10:00 - 12:00	1/TLC	2-11

MSC Form 1910 (Nov 68) FLIGHT PLANNING BRANCH

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
12:00		RECORD BLOCK DATA (TLI +11 HOURS, TLI +25 HOURS, & FLY BY)		VOICE UPDATE: BLOCK DATA
12:30				
13:00 (MSFN)	SLEEP PERIOD (7 HOURS)			
13:30				
14:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	12:00 - 14:00	1/TLC	[illegible]

14:00	MSFN	CDR	CMP	LMP	MCC-H
14:30				BIOMED Sw - RIGHT	
15:00	M S F N	SLEEP PERIOD (7 HOURS			
15:30			EAT PERIOD	EAT PERIOD	
16:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	14:00 - 16:00	1/TLC	2-13

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

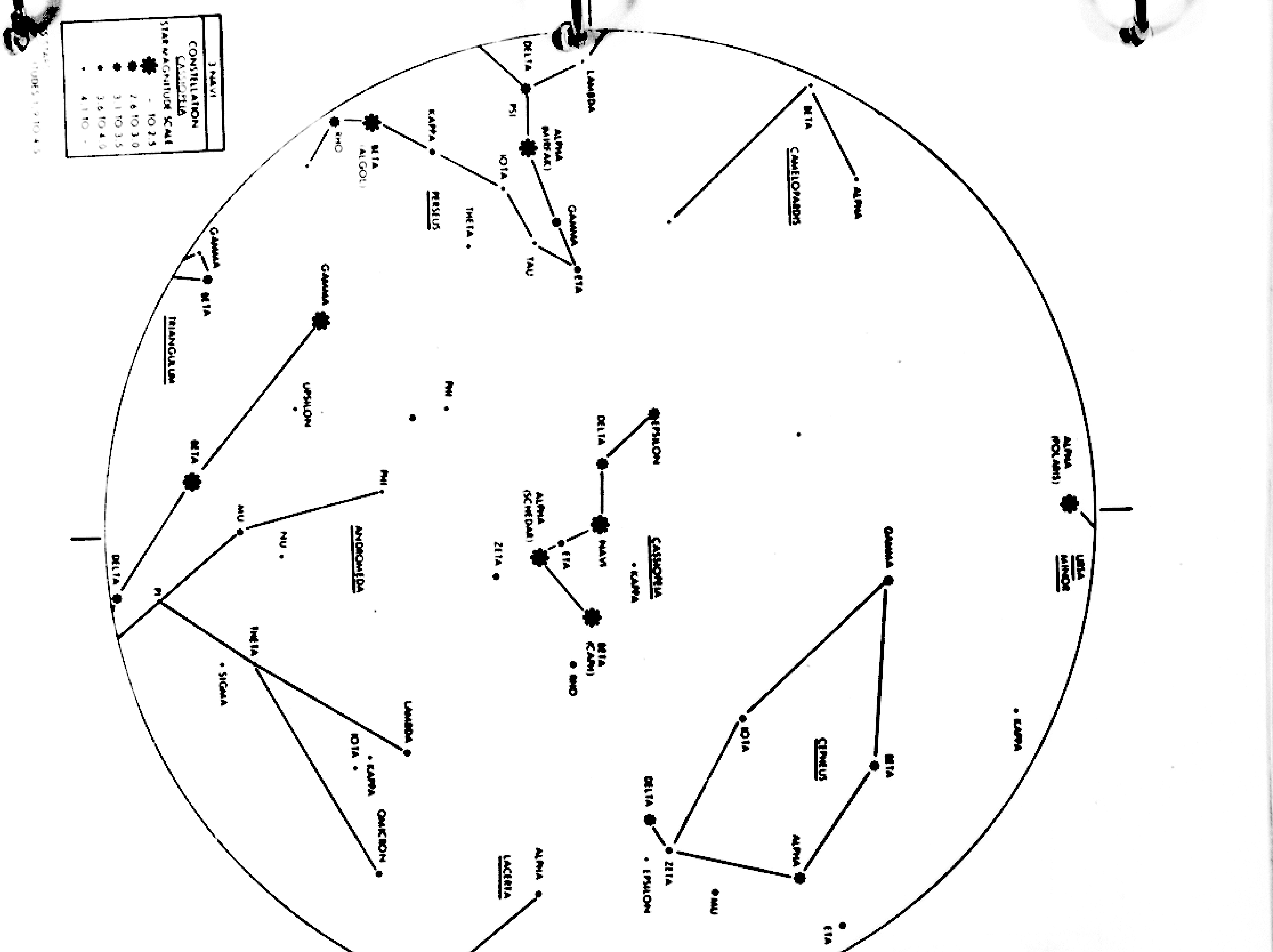
FLIGHT PLAN

16:00	MSFN	CDR	CMP	LMP	MCC-H
16:15					
16:30	M S F N	SLEEP PERIOD (7 HOURS)		MNVR TO P52 ATT	
			IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID 0 3, _ _ STAR ANGLE DIFF _ . _ _		
16:45			TORQUE ANGLES: X _ _ . _ _ _ Y _ _ . _ _ _ Z _ _ . _ _ _		
			SCT STAR VISIBILITY 120° SUN ON NAVI	GDC ALIGN TO IMU MNVR TO SIGHTING ATT	
17:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

Time		CDR	CMP	LMP	MCC-H
17:00	M S F N	SLEEP PERIOD (7 HOURS)			
			TRN BIAS		
17:15					
			CISLUNAR NAVIGATION P23		
			1. STAR ~~22~~ 16 EFH 00120 STAR__ __ E__H 3 SETS		
17:30					
			2. STAR ~~16~~ 22 EFH 00120 STAR__ __ E__H 2 SETS		
17:45			GROUND TRACK DET P21		
18:00				BIOMED Sw - LEFT	

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	17:00 - 18:00	1/TLC	2-15



FLIGHT PLAN

Time		CDR	CMP	LMP	MCC-H
18:00	MSFN	MNVR TO PTC ATT P ~~331~~ 56 Y ~~331~~ 340 P Y ROLL 0.1°/SEC	LiOH CANISTER CHANGE (CARTRIDGE NO. 4 FROM ~~A3~~ B5 INTO CANISTER B)		
18:30					
19:00		EAT PERIOD	SLEEP PERIOD (7 HOURS)	SLEEP PERIOD (7 HOURS)	
19:30		CDR CREW STATUS REPORT			
20:00					

	CDR	CMP	LMP	MCC-H
20:00				
20:30				
21:00		SLEEP PERIOD (7 HOURS)	SLEEP PERIOD (7 HOURS)	
21:30				
22:00				

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	20:00 - 22:00	1/TLC	2-17

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
22:00				
22:30				
23:00		SLEEP PERIOD (7 HOURS)	SLEEP PERIOD (7 HOURS)	
23:30				
24:00				

MSFN

	CDR	CMP	LMP	MCC-H
24:00		SLEEP PERIOD (7 HOURS)	SLEEP PERIOD (7 HOURS)	
24:30				
25:00	EAT PERIOD	RECORD BLOCK DATA (TLI + 25 HOURS, & TLI + 35 HOURS, & FLY BY)	BIOMED Sw - CENTER	VOICE UPDATE: BLOCK DATA
25:30		EAT PERIOD	EAT PERIOD	
26:00	MNVR TO P52 ATT	Chlorinate Water		

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	24:00 - 26:00	2/TLC	2-19

FLIGHT PLAN

00:00
LIFT-OFF
SYSTEMS CK
P52

GO/NO GO FOR TLI
TLI

SEP FROM S-IVB
P52
CMP - STAR/EARTH
HORIZON SIGHTINGS
DOFF & STOW
PGA's (ALL)
PTC

06:00
EAT
PERIOD (ALL)

P52
TLI+6 Hrs
MCC_1
CMP - STAR/EARTH
LANDMARK SIGHTINGS
PTC

EAT
PERIOD (CDR)

12:00

12:00
CDR
SLEEP
PERIOD
(7 HOURS)

CMP & LMP
EAT
PERIOD

P52
CMP - STAR/EARTH
HORIZON SIGHTINGS
PTC

18:00
CDR
EAT
PERIOD

CMP
&
LMP
SLEEP
PERIOD
(7 HOURS)

24:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
[illegible]	[illegible]	November 22, 1968	00:00 -24:00	1	5-1

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
26:00		IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID __ __ __ STAR ANGLE DIFF __.__ __ TORQUE ANGLES: X __ __.__ __ __ Y __ __.__ __ __ Z __ __.__ __ __	CMP/LMP CREW STATUS REPORT	
26:15	GDC ALIGN TO IMU MNVR TO SIGHTING ATTITUDE	TRN BIAS		
26:30 (MSFN)		CISLUNAR NAVIGATION P23 1. STAR 16 EFH 00120 STAR__ __ E__H 1 SET		
26:45		2. STAR 22 EFH 00120 STAR__ __ E__H 1 SET 3. STAR 26 ENH 00110 STAR__ __E__H 1 SET	RECORD MNVR PAD	VOICE UPDATE: MNVR PAD
27:00				

BURN STATUS REPORT

X X ☐	:	ΔTIG
X X	:	BT
☐	.	V_{gx}
TRIM		
X X X		R
X X X		P
X X X		Y
☐	.	V_{gx}
☐	.	V_{gy}
☐	.	V_{gz}
☐	.	ΔV_c
X X X		FUEL
X X X		OX
X X X		UNBALANCE

REMARKS:

MCC'S

BURN CHART

	P or Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
MCC(ALL)	10°/SEC TAKEOVER	10° TAKEOVER	B/T +1 SEC	TRIM TO 0.2 fps

FLIGHT PLAN

TIME		CDR	CMP	LMP	MCC-H
27:00			GROUND TRACK DET P21		P27 UPDATE: STATE VECTOR TGT LOAD
27:15		V47 TRANS LM STATE VECTOR TO CSM SLOT EXT ΔV P30			
27:30	MSFN	SPS/RCS THRUST P40/41 MNVR TO BURN ATT			PIPA BIAS CK
			SXT STAR CK TRANS TO COUCH		
27:45		EMS ΔV TEST			
TLI + 25 HRS		GDC ALIGN	SM RCS MON CK		
28:00		MCC_2 ΔV=NOMINALLY ZERO			

Time		CDR	CMP	LMP	MCC-H
28:00	MSFN	V66 TRANS CSM STATE VECTOR TO LM SLOT MNVR TO SIGHTING ATT MCC_2 BURN STATUS REPORT	SM RCS MONITOR CK TRN BIAS	SPS MONITOR CK ~~INITIATE BAT CHARGE~~ BIOMED Sw - RIGHT	
28:15			CISLUNAR NAVIGATION P23 1. STAR 16 EFH 00120 STAR__ __ E__H 1 SET		
28:30			2. STAR 22 EFH 00120 STAR__ __ E__H 1 SET 3. STAR 21 EFH 00120 STAR__ __ E__H 1 SET		
28:45			4. STAR 26 ENH 00110 STAR__ __ E__H 1 SET GROUND TRACK DET P21	MNVR TO PTC ATT P ~~331~~ Y ~~331~~ P 56 Y 340 ROLL 0.1°/SEC	
29:00		CDR CREW STATUS REPORT			

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	28:00 - 29:00	2/TLC	2-22

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

Time		CDR	CMP	LMP	MCC-H
29:00	MSFN	SLEEP PERIOD (7 HOURS)	LiOH CANISTER CHG (CARTRIDGE NO. 5 FROM ~~A3~~ A5 INTO CANISTER A)		COMM TEST MODE: ~~A~~ ~~5~~ ~~6~~ ~~8~~
29:30				(R,V,R,V, HBR) (R,U. R,V, HBR) (R,V. R,V,LBR) (UVB. R,V, HBR) (V,U.LBR) (R.R) (V,U. DVB,LBR) (V,U. DVB)	4.2 5.2 4.3 8.2 ~~1.0~~ 3.5 ~~0.6~~ 1.7 3.8 3.10
30:00			EAT PERIOD	EAT PERIOD	
30:30				UNSTOW TV CAMERA	
31:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

	CDR	CMP	LMP	MCC-H
31:00			BIOMED Sw - CENTER	
			TV	
31:30			STOW TV CAMERA	
32:00 MSFN	SLEEP PERIOD (7 HOURS)			
32:30				
33:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	31:00 - 33:00	2/TLC	2-24

MSC Form 1910 (Nov 68)

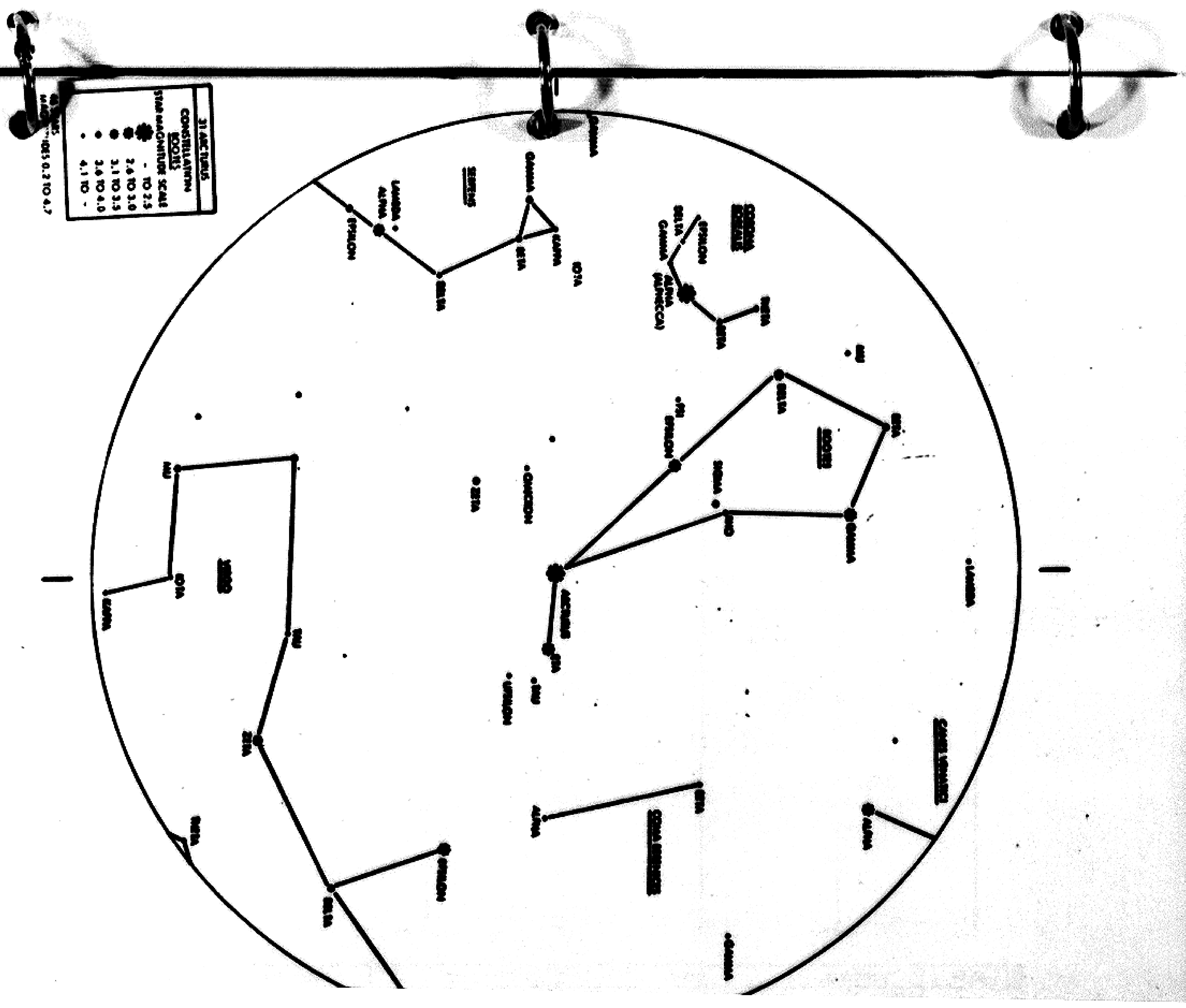
FLIGHT PLANNING BRANCH

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
33:00				
33:15				
33:30 MSFN	SLEEP PERIOD (7 HOURS)			
33:45			MNVR TO P52 ATT	
		IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID 3 1, __ STAR ANGLE DIFF __ — — —.— — (cont'd)		
34:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

	CDR	CMP	LMP	
34:00		TORQUE ANGLES: X __ __.__ __ __ Y __ __.__ __ __ Z __ __.__ __ __	BIOMED Sw - RIGHT	
		SCT STAR VISIBILITY TRN BIAS (70° SUN ON ACTURUS)	MNVR TO SIGHTING ATT	
34:15		**CISLUNAR NAVIGATION P23**		
		1. STAR 16 EFH 00120 STAR__ __ E__H 1 SET		
34:30	MSFN — SLEEP PERIOD (7 HOURS)	2. STAR 22 EFH 00120 STAR__ __ E__H 1 SET		
		3. STAR 26 ENH 00110 STAR__ __ E__H 1 SET		
34:45		GROUND TRACK DET P21	MNVR TO PTC ATT P ~~331~~ 56 Y ~~331~~ 340 P Y ROLL 0.1°/SEC	
		RECORD BLOCK DATA (TLI + 35 HOURS, TLI + 44 HOURS, AND FLY BY)		VOICE UPDATE: BLOCK DATA
35:00				



FLIGHT PLAN

GET		CDR	CMP	LMP	MCC-H
35:00	MSFN				
35:30		SLEEP PERIOD (7 HOURS)	EAT PERIOD	EAT PERIOD	
36:00				CMP/LMP CREW STATUS REPORT	
				BIOMED Sw - LEFT	
36:30		EAT PERIOD	SLEEP PERIOD (7 HOURS)	SLEEP PERIOD (7 HOURS)	
37:00					

CDR CMP LMP

37:00 CDR CREW STATUS REPORT

37:30

M S F N

SLEEP PERIOD (7 HOURS)

SLEEP PERIOD (7 HOURS)

38:00

38:30

39:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	37:00 - 39:00	2/TLC	2-28

FLIGHT PLAN

CDR CMP LMP MCC-H

39:00

39:30

M S F N

SLEEP PERIOD (7 HOURS)

SLEEP PERIOD (7 HOURS)

40:00

40:30

LiOH CANISTER CHANGE (CARTRIDGE NO. 6 FROM ~~A5~~ B5 INTO CANISTER B)

41:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

	CDR	CMP	LMP	
41:00				
41:30				
42:00 MSFN		SLEEP PERIOD (7 HOURS)	SLEEP PERIOD (7 HOURS)	
42:30				
43:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	41:00 - 43:00	2/TLC	2-30

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
43:00			BIOMED Sw - CENTER	
43:30 MSFN	EAT PERIOD	EAT PERIOD	EAT PERIOD	
44:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

44:00

	CDR	CMP	LMP	MCC-H
44:00		RECORD BLOCK DATA (TLI + 44 HOURS, FLY BY, ~~PC + 2 (QUICK RETURN),~~ & PC + 2) ASSUMING FLYBY	CMP/LMP CREW STATUS REPORT	VOICE UPDATE: BLOCK DATA
44:30 MSFN	MNVR TO P52 ATT	IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID __ __ STAR ANGLE DIFF __ __ . __ __ TORQUE ANGLES: X __ __ __ . __ __ __ Y __ __ __ . __ __ __ Z __ __ __ . __ __ __		
45:00	GDC ALIGN TO IMU			

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	44:00 - 45:00	2/TLC	2-32

MSC Form 1810 (Nov 68)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
45:00	MNVR TO SIGHTING ATT	TRN BIAS CISLUNAR NAVIGATION P23		
		1. STAR 33 LNH 00210 STAR __ __ L __ H 3 SETS		
45:15				
45:30 MSFN		2. STAR ~~37 LNH~~ 34 LFH 00280 STAR __ __ L __ H 1 SET		
45:45		3. STAR ~~45 LFH 00220~~ 40 LNH 00210 STAR __ __ L __ H 1 SET		
		~~4. STAR 42 LFH 00240 STAR __ __ L __ H 1 SET~~		
46:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

BURN STATUS REPORT

X X ☐	:	ΔTIG
X X	:	BT
☐	.	V_{gx}
TRIM		
X X X		R
X X X		P
X X X		Y
☐	.	V_{gx}
☐	.	V_{gy}
☐	.	V_{gz}
☐	.	ΔV_c
X X X		FUEL
X X X		OX
X X X		UNBALANCE

REMARKS:

2-55a

MCC'S

BURN CHART

	P or Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
MCC(ALL)	10°/SEC TAKEOVER	10° TAKEOVER	B/T +1 SEC	TRIM TO 0.2 fps

FLIGHT PLAN

		CDR	CMP	LMP	MCC-H
46:00				BIOMED Sw - RIGHT	
			GROUND TRACK DET P21	RECORD MNVR PAD	P27 UPDATE: STATE VECTOR TGT LOAD VOICE UPDATE: MNVR PAD
46:15		V47 TRANS LM STATE VECTOR TO CSM SLOT → EXT ΔV P30			
46:30	MSFN	SPS/RCS THRUST P40/41			
		MNVR TO BURN ATT	SXT STAR CK		PIPA BIAS CK
			TRANS TO COUCH		
46:45		EMS ΔV TEST			
		GDC ALIGN TO IMU	SM RCS MON CK		
LOI -22 HRS 47:00		$\boxed{\text{MCC}_3\ \Delta V\text{=NOMINALLY ZERO}}$			

TIME	CDR	CMP	LMP	
47:00	MNVR TO SIGHTING ATT	SM RCS MON CK	SPS MON CK ~~INITIATE BAT CHARGE~~	
	V66 TRANS CSM STATE VECTOR TO LM SLOT →			
	MCC_3 BURN STATUS REPORT	TRN BIAS		
47:15		**CISLUNAR NAVIGATION P23**		
		1. STAR 16 EFH 00:20 STAR__ __ E__H 1 SET		
47:30 MSFN		2. STAR 22 EFH 00:20 STAR__ __ E__H 1 SET		
		3. STAR 26 ENH 00:10 STAR__ __ E__H 1 SET		
47:45	MNVR TO PTC ATT P ~~331~~ Y ~~331~~ P 36 Y 340 ROLL 0.1°/SEC	GROUND TRACK DET P21		
48:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	47:00 - 48:00	2/TLC	2-35

FLIGHT PLAN

TIME			
24:00			
		EAT PERIOD (ALL)	
	P52 CMP - STAR/EARTH HORIZON SIGHTINGS		
TLI + 25 Hrs	$\boxed{MCC_2}$		
	CMP - STAR/EARTH HORIZON SIGHTINGS PTC		
30:00		CMP & LMP EAT PERIOD	
	TV		CDR SLEEP PERIOD (7 HOURS)
	P52 CMP - STAR/EARTH HORIZON SIGHTINGS PTC		
36:00		CMP & LMP EAT PERIOD	

TIME			
36:00		CDR EAT PERIOD	CMP & LMP SLEEP PERIOD (7 HOURS)
42:00			
		EAT PERIOD (ALL)	
	P52 CMP - STAR/LUNAR HORIZON SIGHTINGS		
LOI -22 Hr	$\boxed{MCC_3}$ CMP - STAR/EARTH HORIZON SIGHTINGS		
48:00			

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
		[illegible] 1968	24:00 - 48:00	2	5-2

FLIGHT PLAN

	MSFN	CDR	CMP	LMP	MCC-H
48:00					
48:30					
49:00	M S F N			BIOMED Sw - CENTER	
49:30		EAT PERIOD	EAT PERIOD	EAT PERIOD	
50:00					

Time		CDR	CMP	LMP	MCC-H
50:00			CHLORINATE WATER		
50:30					
51:00	MSFN		LiOH CANISTER CHANGE (CARTRIDGE NO.7 FROM B6 INTO CANISTER A)	RECORD BLOCK DATA (FLY BY & PC + 2 HOURS)	VOICE UPDATE: BLOCK DATA
51:30		MNVR TO P52 ATT	IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID ___ ___ STAR ANGLE DIFF ___ ___ ___.___ ___ (cont'd)		
52:00		CDR CREW STATUS REPORT			

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	50:00 - 52:00	3/TLC	2-37

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

Time		CDR	CMP	LMP	MCC-H
52:00			TORQUE ANGLES: X ___ ___.___ ___ ___ Y ___ ___.___ ___ ___ Z ___ ___.___ ___ ___	BIOMED Sw - RIGHT	
				TRANS TO LH COUCH MNVR TO SIGHTING ATT GDC ALIGN TO IMU	
52:15			TRN BIAS $\boxed{\text{CISLUNAR NAVIGATION P23}}$		
			1. STAR ~~37~~ 33 LNH 00240 STAR___ ___ L__H 3 SETS		
52:30	MSFN	SLEEP PERIOD (7 HOURS)			
52:45			2. STAR ~~33~~ 34 ~~LNH~~ LFH 00220 STAR___ ___ L__H 1 ~~X~~ SET~~S~~		
			3. STAR 40 LNH 00210 STAR ___ L - H 1 SET		
53:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

53:00

53:15 GROUND TRACK DET P21

MNVR TO PTC ATT
P ~~331~~ Y ~~231~~
P 86 Y 340
ROLL 0.1°/SEC

53:30 MSFN

SLEEP PERIOD
(7 HOURS)

53:45

54:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	53:00 - 54:00	3/TLC	2-39

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H

54:00

54:15

54:30 MSFN

SLEEP PERIOD
(7 HOURS)

54:45

55:00

BIOMED Sw - CENTER
UNSTOW TV CAMERA

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

55:00	MSFN	CDR: SLEEP PERIOD (7 HOURS)			
55:15				TV	
55:30				STOW TV CAMERA	
55:45					
56:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	55:00 - 56:00	3/TLC	2-41

FLIGHT PLAN

		CDR	CMP	LMP	MCC-H
56:00	MSFN	SLEEP PERIOD (7 HOURS)	EAT PERIOD	EAT PERIOD	
56:30					
57:00					
57:30					
58:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	56:00 - 58:00	3/TLC	2-42

CDR

58:00

58:30

SLEEP PERIOD
(7 HOURS)

M
S
F
N

59:00

BIOMED Sw - LEFT

59:30

EAT PERIOD

P27 UPDATE:
STATE
VECTOR (CSM + LM)
TGT LOAD
REFSMMAT
VOICE
UPDATE
MNVR PAD + BLOCK DATA

60:00

CDR CREW STATUS REPORT

RECORD MNVR PAD + BLOCK DATA (PC+L, FAST RETURN + PC+2 MIN ΔV ASSUMING MCC4)

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	58:00 - 60:00	3/TLC	2-43

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

BURN STATUS REPORT

X X [ ] : ΔTIG

X X : BT

[ ] . V_{gx}

TRIM

X X X R

X X X P

X X X Y

[ ] . V_{gx}

[ ] . V_{gy}

[ ] . V_{gz}

[ ] . ΔV_c

X X X FUEL

X X X OX

X X X UNBALANCE

REMARKS:

MCC'S

BURN CHART

	P or Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
MCC(ALL)	10°/SEC TAKEOVER	10° TAKEOVER	B/T +1 SEC	TRIM TO 0.2 fps

	CDR	CMP	LMP	MCC-H
60:00	MNVR TO P52 ATT	IMU REALIGN P52 OPTION 1 - PREFERRED STAR ID ___ . ___ STAR ANGLE DIFF ___ . ___ TORQUE ANGLES: X ___ ___ . ___ ___ ___ Y ___ ___ . ___ ___ ___ Z ___ ___ . ___ ___ ___		
60:15				
60:30 MSFN	EXT ΔV P30 SPS/RCS THRUST P40/41 MNVR TO BURN ATT	SXT STAR CK		PIPA BIAS CK
60:45	EMS ΔV TEST	TRANS TO COUCH		
61:00	GDC ALIGN TO IMU $\boxed{\text{MCC}_4\ \Delta\text{V = NOMINALLY ZERO}}$	SM RCS MON CK		

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	60:00 - 61:00	3/TLC	2-44

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
61:00	V66 TRANS CSM STATE VECTOR TO LM SLOT →	SM RCS MON CK GROUND TRACK DET P21 (STATE VECTOR CK)	SPS MON CK ~~INITIATE BAT CHARGE~~ ECS REDUNDANT COMP CK	
	MNVR TO PTC ATT P 122 Y 315 P ___ Y ___			
61:15	ROLL 0.1 °/SEC			
61:30 MSFN				
61:45				
62:00		LiOH CANISTER CHANGE (CARTRIDGE NO.8 FROM B6 INTO CANISTER B)	CMP/LMP CREW STATUS REPORT	

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

CDR

62:00

62:30

M
S
F
N

63:00

SLEEP PERIOD
(6 HOURS)

SLEEP PERIOD
(6 HOURS)

63:30

64:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	62:00 - 64:00	3/TLC	2-46

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

CDR CMP LMP MCC-H

64:00

64:30

M
S
F
N

65:00

SLEEP PERIOD
(6 HOURS)

SLEEP PERIOD
(6 HOURS)

65:30

66:00

MNVR TO P52 ATT

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

66:00 · 66:15 · 66:30 · 66:45 · 67:00

MSFN

CDR

IMU REALIGN P52
OPTION 3 - REFSMMAT
STAR ID ___ ___ ___
STAR ANGLE DIFF ___.___ ___
TORQUE ANGLES:
X ___ ___.___ ___ ___
Y ___ ___.___ ___ ___
Z ___ ___.___ ___ ___

GDC ALIGN TO IMU
~~MNVR TO PTC ATT~~
~~P 122 Y 315~~
~~P ___ Y ___~~
~~ROLL 0.1°/SEC~~

RECORD MNVR PAD FOR LOI₁

CMP

SLEEP PERIOD
(6 HOURS)

LMP

SLEEP PERIOD
(6 HOURS)

MCC-H

VOICE
UPDATE:
MNVR PAD

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	66:00 - 67:00	3/TLC	2-48

MSC Form 1910 (Nov 68) FLIGHT PLANNING BRANCH

FLIGHT PLAN

67:00 · 67:15 · 67:30 · 67:45 · 68:00

MSFN

CDR

MNVR TO LOI_1 ATT
SXT STAR CK

MNVR TO PTC ATT
P 122 Y 315
P ___ Y ___
ROLL 0.1°/SEC

RECORD BLOCK DATA
(TEI_1, TEI_2, & PC+2)
& MAP UPDATE

CMP

SLEEP PERIOD
(6 HOURS)

LMP

SLEEP PERIOD
(6 HOURS)

MCC-H

VOICE
UPDATE:
BLOCK DATA
& MAP
UPDATE 1/2 +
MNVR PAD
P27 UPDATE:
STATE VECTOR
(CSM + LM) +
TGT LOAD

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

	CDR	CMP	LMP	MCC-H
68:00	GO/NO GO FOR LOI_1		CMP/LMP CREW STATUS REPORT	GO/NO GO
		GROUND TRACK DET - P21 (LOI ALTITUDE DET)	~~RECORD MANEUVER PAD~~	~~P27 UPDATE: STATE VECTOR TARGET LOAD~~
	MNVR TO P52 ATT			~~VOICE UPDATE: MNVR PAD~~
		IMU REALIGN P52 OPTION 3 - REFSMMAT AND GYRO DRIFT TEST STAR ID __ __ STAR ANGLE DIFF __.__ TORQUE ANGLES: X __ __.__ __ __ Y __ __.__ __ __ Z __ __.__ __ __		
68:30 (MSFN)			PRE LOI SYSTEMS CKS: C&W CK CM RCS CK SM RCS CK SPS PERIODIC MONITOR EPS PERIODIC MONITOR ECS PERIODIC MONITOR NON ESS BUS - MNB INITIATE WATER BOILER	
	EXTERNAL ΔV P30 EMS CK SPS THRUST P40 MNVR TO BURN ATT	SXT STAR CK TRANSFER TO COUCH		
	GDC ALIGN TO IMU			
68:57				
69:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	68:00 - 69:00	3/TLC	2-50

FLIGHT PLANNING BRANCH

MSC Form 1016 (?) (?)

$h_A = 169.1$
$h_p = 60.5$

BURN STATUS REPORT

X X ☐ :0.T. ΔTIG
X X 4:06.5 BT
☑ 1.4 V_{gx}
— TRIM —
X X X R
X X X NOM P
X X X Y
☑ -1.4 V_{gx}
☐ 0.0 V_{gy}
☑ 0.2 V_{gz}
☑ 20.2 ΔV_c
X X X FUEL
X X X OX
X X X UNBALANCE

REMARKS:

2-50a

LOI_1 BURN CHART

	P or Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
LOI_1	10°/SEC TAKEOVER	10° TAKEOVER	B/T+6 SEC	NO TRIM

LOI_1 ABORT MODES

LOI V_{go}	B/T	TRAJECTORY	ABORT MODE
3050-2100	0 -1:20	HYPERBOLIC	COAST OUT OF SPHERE-P37
2100-1650	1:20-2:00	UNSTABLE	5 HR COAST. MODE I ABORT
1650-0	2:00-4:06	LUNAR ORBIT	MODE III ABORT AFTER 1 REV

69:00

BEGIN REV 1

69:30

70:00

CDR

LOI 1

GDC ALIGN TO IMU/ORB RATE (115 nm)

GO INERTIAL

ROLL RIGHT 180° AT 2°/SEC, CONT INERTIAL ATT HOLD

LOI 1 BURN STATUS REPORT

16/18/NEW C EX, 1/250, SPOT, 1 FPS, (1 MAG)

EAT PERIOD

M S F N

CMP

SM RCS MONITOR CK

GETI = 69:07:30
ΔV = 2991 fps
BT = 4 MIN 5.8 SEC

V66 TRANS CSM STATE VECTOR TO LM SLOT

SM RCS MONITOR CK

2/~~80~~ 160/B3 + 1/~~150~~ 80/A8, R&B FILTER/U4, SPOT/R13, INT/U4, 2 B&W FILM PACKS #A8

UNSTOW ORBITAL CHARTS, 16/18/NEW C-EX/CABLE/BRKT/U3

~~2/80/NEW B&W~~

EAT PERIOD

LMP

~~BIOMED SW~~ CENTER

SPS MONITOR CK

~~INITIATE BAT CHARGE~~

CAMERA PREP

V64 ACQUIRE MSFN ON HGA

2/150/NEW B&W 1/250

1/80/B&W, 1/250

EAT PERIOD

MCC-H

ACQUIRE S/C

DUMP DSE

PIPA BIAS CK

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	69:00 - 70:00	3/LPO 1	2-51

FLIGHT PLAN

70:00
70:02
70:14
70:30
70:55
71:00

CDR

STOP 16MM CAMERA

EAT PERIOD

GDC ALIGN TO IMU

PITCH UP TO ORDEAL 180, 315, 0 AT 0.2°/SEC (HEADS DWN COAS TRACK ATT) BEGIN ORB RATE

M S F N

CMP

EAT PERIOD

IMU REALIGN P52 OPTION 3 - REFSMMAT AND GYRO DRIFT TEST
STAR ID __ __
STAR ANGLE DIFF __ . __ __

TORQUE ANGLES:
X __ __ . __ __ __
Y __ __ . __ __ __
Z __ __ . __ __ __

EAT PERIOD

PREPARE AND REVIEW CNTL POINT AND LDMK CHARTS
1/150/B&W

LMP

1/50/HS, 1/4 F 2.8

VERIFY SYSTEMS STATUS

EAT PERIOD

UNSTOW TV CAMERA & BRKT & TELEPHOTO LENS (A) TEMP STOW INT

BIOMED SW — CENTER

RECORD BLOCK DATA (TEI_1 & TEI_2)

1/80/B&W, 1/250, F 2.8, INT

2/150/B&W, 1/250, F 2.8

16/18/~~NEW~~ C EX, 1/250, F 2.8, 1 fps (1/3 MAG)

MCC-H

VOICE UPDATE: BLOCK DATA

GO FOR REV 2 ~~(?)~~!

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

	CDR	CMP	LMP	MCC-H
71:00		CONTROL PT (CP) SIGHTING (THRU HATCH WINDOW)		
	MAINTAIN ORB RATE AND BEGIN COAS GROUND TRACK DETERMINATION		START 16 MM CAMERA AT TERMINATOR	
71:09		CP1 7 1 : 1 3 : 4 6 TCA 7 1 : 1 4 : 5 7 CP1 NO. 504 LAT -05.250°S LONG -162.700°W ALT 000.00 nm	SPOT METER READINGS PHOTOGRAPH TARGETS OF OPPORTUNITY	
BEGIN REV 2		CP2 7 1 : 2 7 : 4 0 TCA 7 1 : 2 8 : 3 2 CP2 NO. 526 LAT -10.200°S LONG +155.100°E ALT 000.00 nm	CONFIGURE FOR TV	
71:30	YAW RT 45° FOR TV ORDEAL 180, 315, ~~45~~ 315°			
71:38	TV	CP3 7 1 : 4 6 : 3 8 TCA 7 1 : 4 7 : 2 6 CP3 NO. 334 LAT -09.000°S LONG ~~+95.900°E~~ +96.350°E ALT 000.00 nm	V64 ACQUIRE MSFN ON HGA	ACQUIRE S/C P27 UPDATE: STATE VECTOR (CSM + LM) TARGET LOAD
MSFN	YAW LEFT 45° TO ORDEAL 180, 315, 0	PSEUDO LDG SITE LDMK SIGHTING (THRU HATCH WINDOW)	RECORD MAP UPDATE	VOICE UPDATE: MAP UPDATE 2/3
72:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	71:00 - 72:00	3/LPO 1+2	2-53

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
72:00	MAINTAIN ORBITAL RATE AND COAS GROUND TRACK DETERMINATION	B-1 7 2 : 0 9 : 1 0 TCA 7 2 : 0 9 : 4 2 LDMK NO. B1 LAT +02.675°N LONG +35.025°E ALT -000.99 nm		
72:10	GO/NO GO FOR LOI_2	LiOH CANISTER CHANGE (CARTRIDGE 9 FROM B6 INTO CANISTER A)	BIOMED Sw - LEFT STOP 16 MM CAMERA RECORD MNVR PAD, BLOCK DATA (TEI_3, TEI_3 NO LOI_2).	GO/NO GO VOICE UPDATE: MNVR PAD BLOCK DATA
72:23	GO INERTIAL		Cryo fan cycle	
72:30 MSFN		IMU REALIGN P52 OPTION 3 - REFSMMAT AND GYRO DRIFT TEST STAR ID ___ ___ STAR ANGLE DIF. ___ ___ TORQUE ANGLES: X ___ ___.___ ___ ___ Y ___ ___.___ ___ ___ Z ___ ___.___ ___ ___	1/80/HS, 1/4, F 2.8 PRE LOI SYSTEMS CKS: C&W CK CM RCS CK SPS PERIODIC MONITOR EPS PERIODIC MONITOR ECS PERIODIC MONITOR	DUMP DSE PIPA BIAS CK
	EXTERNAL ΔV P30 EMS CK SPS THRUST - P40 MNVR TO BURN ATT GO INERTIAL	CHLORINATE WATER Put off til later		
73:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
			72:00 - 73:00	4/LPO 3	2-54

2-54a

BURN STATUS REPORT

X X [ ]	:0.7	ΔTIG
X X	11:60	BT
[+]	.2	V_{gx}

TRIM

X X X		R
X X X		P
X X X		Y
[ ]	.2	V_{gx}
[ ]	1.8	V_{gy}
[ ]	.2	V_{gz}
[ ]	9.4	ΔV_c
X X X	/	FUEL
X X X	/	OX
X X X	/	UNBALANCE

REMARKS:

Ha 62.0
Hp 60.8

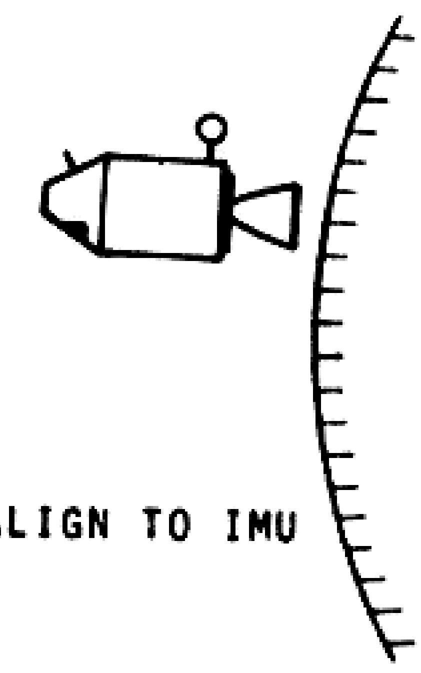
LOI$_2$

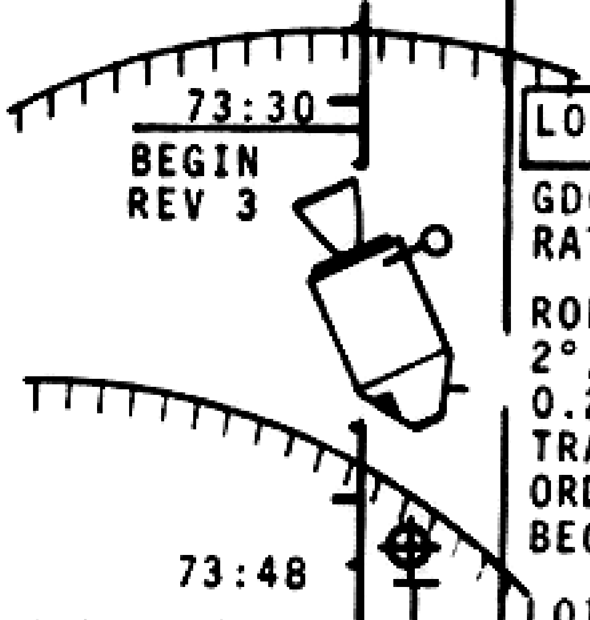
BURN CHART

	P or Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
LOI$_2$	10°/SEC TAKEOVER	10° TAKEOVER	B/T +1 SEC	NO TRIM

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
73:00			2/80/B&W, 1/250	
73:03	~~MNVR TO BURN ATT GO INERTIAL~~	SXT STAR CK	16/18/C EX, 1/250 ~~f8~~ 1/2 MAG F CHART, 6 fps, BRACK	
73:09		TRANSFER TO COUCH	1/150/B&W, ~~1/60~~, 1/250 f8 CHART, BRKT	
			PRE LOI SYSTEMS CKS	
73:17				
	GDC ALIGN TO IMU	SM RCS MONITOR CK		
		GETI = 73:30:54 ΔV = 138.5 fps BT = 9.7 SEC		
73:30 BEGIN REV 3	LOI$_2$ GDC ALIGN TO IMU/ORB RATE (60 nm)	V66 TRANS CSM STATE VECTOR TO LM SLOT	SPS MONITOR CK	
	ROLL RIGHT 180° AT 2°/SEC PITCH DWN AT 0.2°/SEC (HEADS DWN TRACKING ATT) ORDEAL 180, 308, 0 BEGIN ORB RATE	SM RCS MONITOR CK	INITIATE BAT B CHARGE	
73:48	LOI$_2$ BURN STATUS REPORT	REST PERIOD (2 HOURS)	V64 ACQUIRE MSFN ON HGA	ACQUIRE S/C PIPA BIAS CK
74:00			BIOMED SW - RIGHT	





Green starts

MSFN

74:00

CDR

TRAINING PHOTOGRAPHY

PITCH UP TO TRACK LDMK

7 4 : 1 5 : __ __

__ __ : __ __ : __ __

74:17

STOP CAMERA

PITCH DWN TO HEADS DWN
TO REACQUIRE MSFN
ORDEAL 180, 0, 0
GO INERTIAL

74:30

M
S
F
N

INSTALL 16 MM CAMERA

GDC ALIGN TO IMU

75:00

CMP

T_0 = 8:1 TCA

	F16	F70	
T_0 -7	8	8	
-4	5.6	5.6	
-2	4	—	
-1	2.8	—	1/60
0	2	—	—

REST PERIOD
(2 HOURS)

LMP

TRAINING PHOTOGRAPHY
70mm & 16mm

START CAMERAS 7 MIN PRIOR
LDMK, 70mm - 20 SEC INT
UNTIL TRACKING - THEN
5 SEC, F STOP FROM CHART

STOP CAMERAS

1/80/HS, 1/4, F 2.8

VERIFY SYSTEMS STATUS

RECORD MAP UPDATE

IMU REALIGN P52
OPTION 3 - REFSMMAT
AND GYRO DRIFT TEST
STAR ID __ __ __
STAR ANGLE DIFF __ . __ __

TORQUE ANGLES:
X __ __ __ . __ __ __
Y __ __ __ . __ __ __
Z __ __ __ . __ __ __

RECORD BLOCK DATA
(TEI$_3$ & TEI$_4$)
GEGENSCHEIN/LIBRATION PT

MCC-H

VOICE
UPDATE:
MAP UPDATE
3/4

DUMP DSE

VOICE
UPDATE:
BLOCK DATA

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	74:00 - 75:00	4/LPO 3	2-56

FLIGHT PLAN

75:00

CDR

PITCH UP TO VERTICAL AT
0.2°/SEC (+Z FWD)
ORDEAL 180, 270, 0.
BEGIN ORB RATE

75:09

16/18/NEW C EX/1 fps
1/250, F 2, 1 fps
START CAMERA S/S SR +
6 MIN

75:15

75:30

BEGIN
REV 4

SUBSOLAR POINT

75:46 __ __ : __ __

ROLL RIGHT 180° AT
2°/SEC (-Z FWD) + PITCH DWN 20°
MAINTAIN ORB RATE

75:46

M
S
F
N

76:00

CMP

REST PERIOD
(2 HOURS)

INSTALL SXT BRKT &
16mm CAMERA. TAKE
PHOTOS AS TIME PERMITS.

SA 0° TA 0°
16/SXT/CEX 1/2 MAG
1/60 6FPS

LMP

1/80/NEW B&W, 1/250,
F 2.8 BRKT/INT

2/150/B&W

START CAMERA 1 S/C
SR + 6 MIN

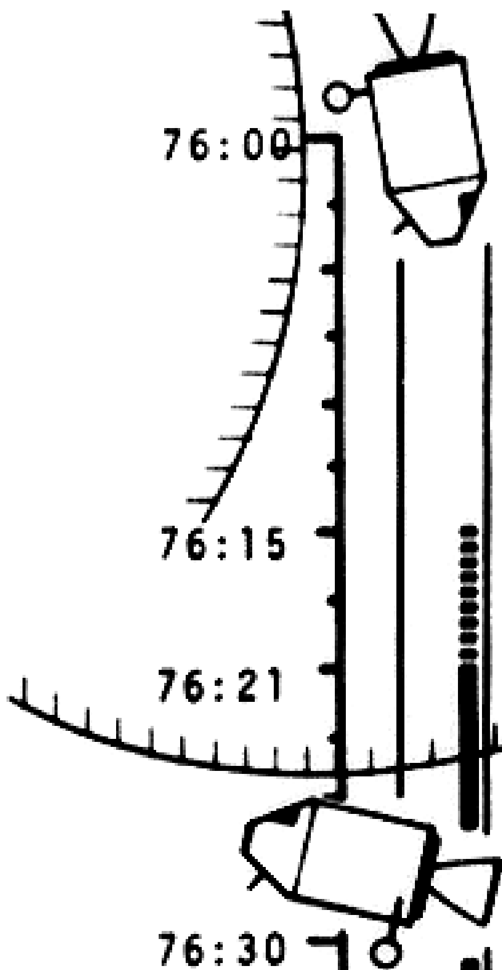
VERTICAL STEREO
PHOTOGRAPHY
EXTRA EXP EACH
5 MIN - NOTE TIME

ACQUIRE
MSFN ON
OMNI ANT

MCC-H

ACQUIRE S/C

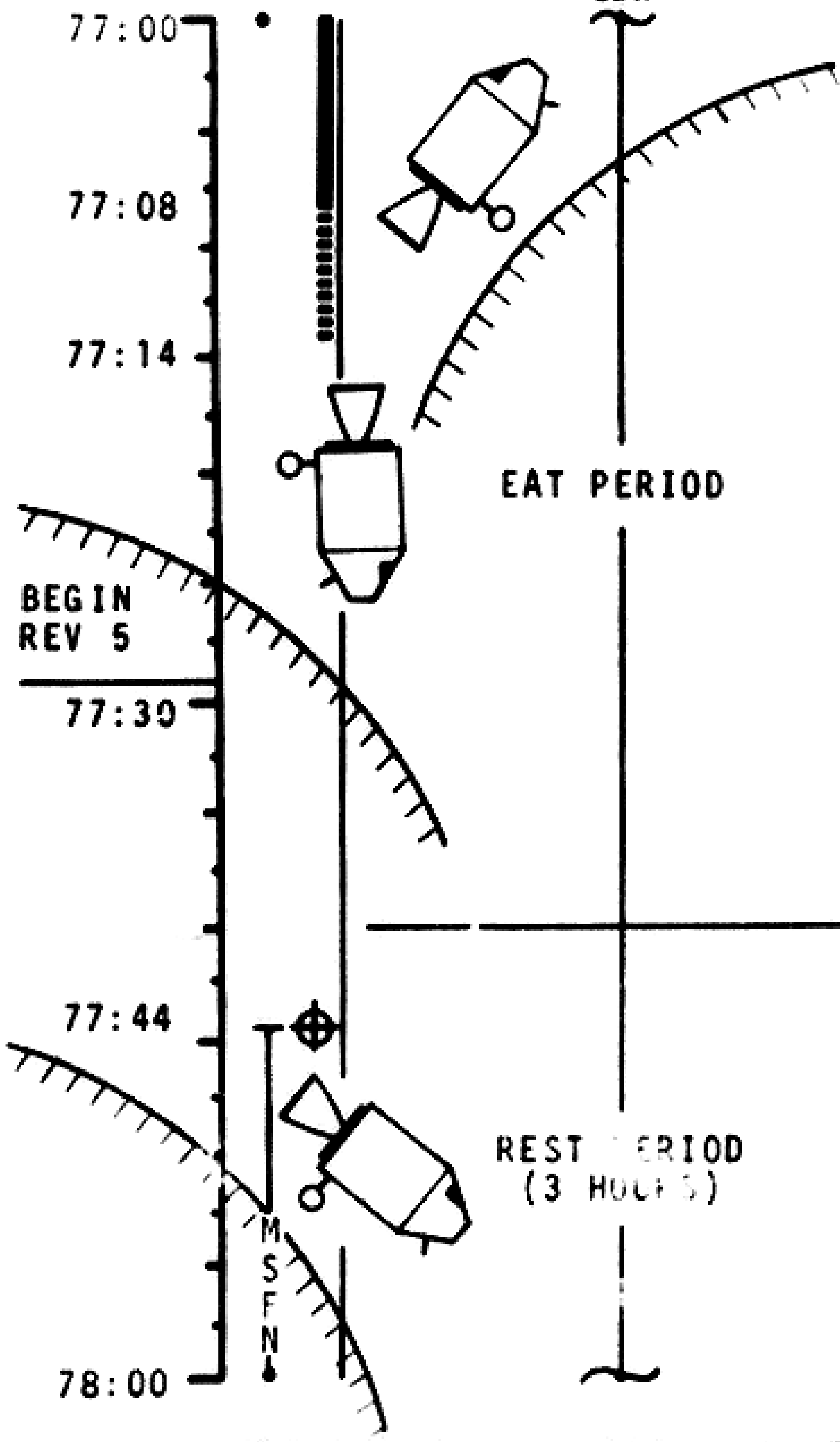
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	75:00 - 76:00	4/LPO 3+4	2-57



TIME	CDR	CMP	LMP	MCC-H
76:00	PITCH UP AT 0.2°/SEC TO 0, 355, 0 BEGIN ORB RATE STOP CAMERAS AT TERMINATOR	IPI ACQ __ __:__ __:__ __ IPII ACQ __ __:__ __:__ __ LDMK LIGHTING EVALUATION (THRU OPTICS) OF LDMK B1 *START CAMERA IPI-2MIN*	AT PITCH UP 1/80/B&W, 1/250, F 11 STOP CAMERA 1 AT TERM RECORD MAP UPDATE	VOICE UPDATE: MAP UPDATE 4/5 P27 UPDATE: STATE VECTOR
76:15				
76:21		*V47 TRANS LM STATE VECTOR TO CSM SLOT*		
76:30	ROLL RIGHT 180° AT 2°/SEC (HEADS DWN) GO INERTIAL MSFN	IMU REALIGN P52 OPTION 3 - REFSMMAT AND GYRO DRIFT TEST STAR ID __ __ STAR ANGLE DIFF __ __ . __ __ TORQUE ANGLES: X __ __:__ __ __ Y __ __:__ __ __ Z __ __:__ __ __	V64 ACQUIRE MSFN ON HGA 1/80/HS, 1/4, F 2.8 (20 SEC STRIP - 20 EXP) *NO INT, COPERNICUS H/S* VERIFY SYSTEMS STATUS RECORD BLOCK DATA: (~~TEI_4~~ & TEI_5) TRANS TO LEFT COUCH	ACQUIRE S/C DUMP DSE VOICE UPDATE: BLOCK DATA
	EAT PERIOD GDC ALIGN TO IMU		1/80/B&W, 1/250, F 2.8 INT (CDR) BIOMED Sw - CENTER	
76:58 77:00				

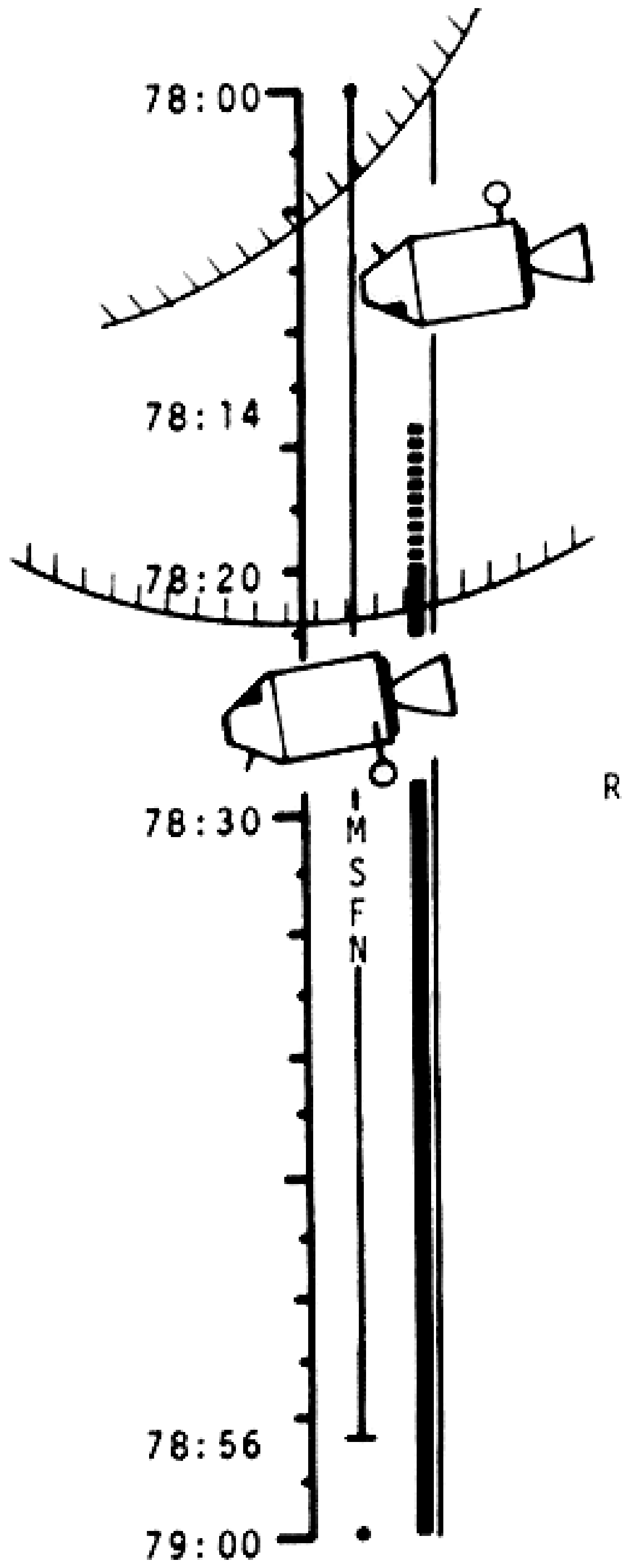
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	76:00 - 77:00	4/LPO ~~4~~	2-58

FLIGHT PLAN



TIME	CDR	CMP	LMP	MCC-H
77:00 77:08 77:14		MAP REVIEW *16/SXT/CEX 1/3 MAG 1/250 6FPS* P22 MANUAL ACQ SA 0°, TA 10°	ROLL LEFT 180° AT 2°/SEC PITCH DWN AT 0.2°/SEC TO ORDEAL 0, 5, 0. BEGIN ORB RATE. *~~LIMB BRIGHTENING OBS~~* 2/150/B&W, 1/250 F 2.8	
BEGIN REV 5 77:30	EAT PERIOD	CNTL POINT SIGHTING *START CAMERA IPI ACQ* IPI ACQ *77:29:42* CP2 NO. 526 LAT -10.200°S LONG/2 +77.550° ALT +000.00 nm NEW COORDINATES LAT __ __ . __ __ __° LONG/2 __ __ . __ __ __° ALT __ __ __ . __ __ nm	2/150/B&W, 1/250 (CDR) PITCH DWN AT 0.3°/SEC AT ACQ	
77:44	REST PERIOD (3 HOURS) MSFN	REVIEW LDMK MAP *16/SXT/CEX 1/3 MAG 1/60 6FPS* *START CAMERA IPI TCA*	ACQUIRE MSFN ON OMNIS AT LOSS PITCH UP AT 0.2°/SEC TO ORDEAL 0, 355, 0. BEGIN ORB RATE. REPORT NEW CP COORDINATES	ACQUIRE S/C
78:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE



CDR

REST PERIOD
(3 HOURS)

CMP

PSEUDO LDG SITE SIGHTINGS
P22 - AUTO OPTICS
IPI ~~77 : 39 : 46~~
TCA 78 : 10 : 35
LDMK NO. B1
LAT +02.675°N
LONG/2 +17.512°
ALT -000.99 nm
NEW COORDINATES
LAT +02.515°
LONG/2 +17.665°
ALT -_ _ _ .02 nm

V47 TRANS LM STATE VECTOR TO CSM SLOT

IMU REALIGN P52
OPTION 3 - REFSMMAT
AND GYRO DRIFT TEST
STAR ID __ __, __ __
STAR ANGLE DIFF __ __

TORQUE ANGLES:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __

LMP

PITCH DWN 0.3°/SEC AT ACQ
AT LOSS ROLL RIGHT 180°
AT 2°/SEC TO HEADS DWN
GO INERTIAL

V64 ACQUIRE MSFN ON HGA
1/80/HS, 1/4, F 2.8

REPORT NEW COORDINATES
RECORD BLOCK DATA
(TEI_6), & MAP UPDATE
VERIFY SYSTEMS STATUS

SC EXT ATM (EARTHSHINE)
1/80/HS F2.8
B 65SEC, B 15SEC, 1/60
90° FROM SUN

GDC ALIGN TO IMU
~~16/18/ C EX, 1/250, 1 fps~~
~~(__ MAGS)~~
2/150/C121/1/250 SPOT
GENERAL OBSERVATIONS
SC EXT ATM (DARKNESS)

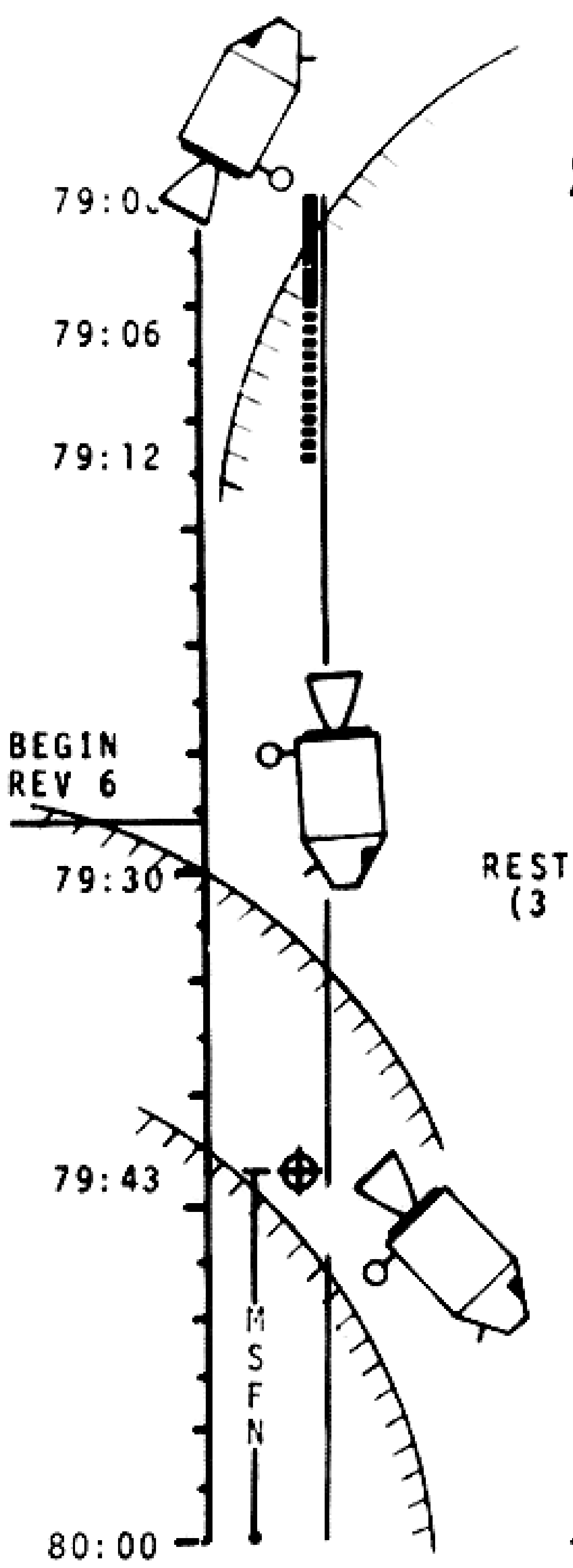
MCC-H

ACQUIRE S/C
P27 UPDATE:
STATE VECTOR
VOICE UPDATE:
BLOCK DATA & MAP UPDATE
5/6

DUMP DSE

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	78:00 - 79:00	4/LPO 5	2-60

FLIGHT PLAN



CDR

1,2 auto good
works well

REST PERIOD
(3 HOURS)

CMP

REVIEW CONTROL POINT MAP

16/SXT/CEX 1/3 MAG
1/500 1FPS

CNTL POINT SIGHTING
START CAMERA IPI TCA

P22 - AUTO OPTICS
IPI ~~79 : 10 : 53~~
TCA __ __ : __ __ : __ __
CP2 NO. 526
LAT -10.200°S
LONG/2 +77.550°
ALT +000.00 nm

NEW COORDINATES
LAT __ __.__ __ __°
LONG/2 __ __.__ __ __°
ALT __ __ __.__ __ nm

REVIEW LDMK MAP
CP3 New Coord.
08766
48631

PSEUDO LDG SITE SIGHTING
START CAMERA IPI TCA
1/60 1FPS

LMP

ROLL 180° AT 2°/SEC
PITCH DWN AT 0.2°/SEC TO
OPDEAL 0, 5, 0.
BEGIN ORB RATE
LIMB BRIGHTENING
1/80/HS 1/60 F2.8
SR-15SEC TO SR

SC EXT ATM (SUNLIT)

PITCH DWN 0.3°/SEC AT ACQ

AT LOSS PITCH UP AT
0.2°/SEC TO
ORDEAL 0, 355, 0. BEGIN
ORB RATE

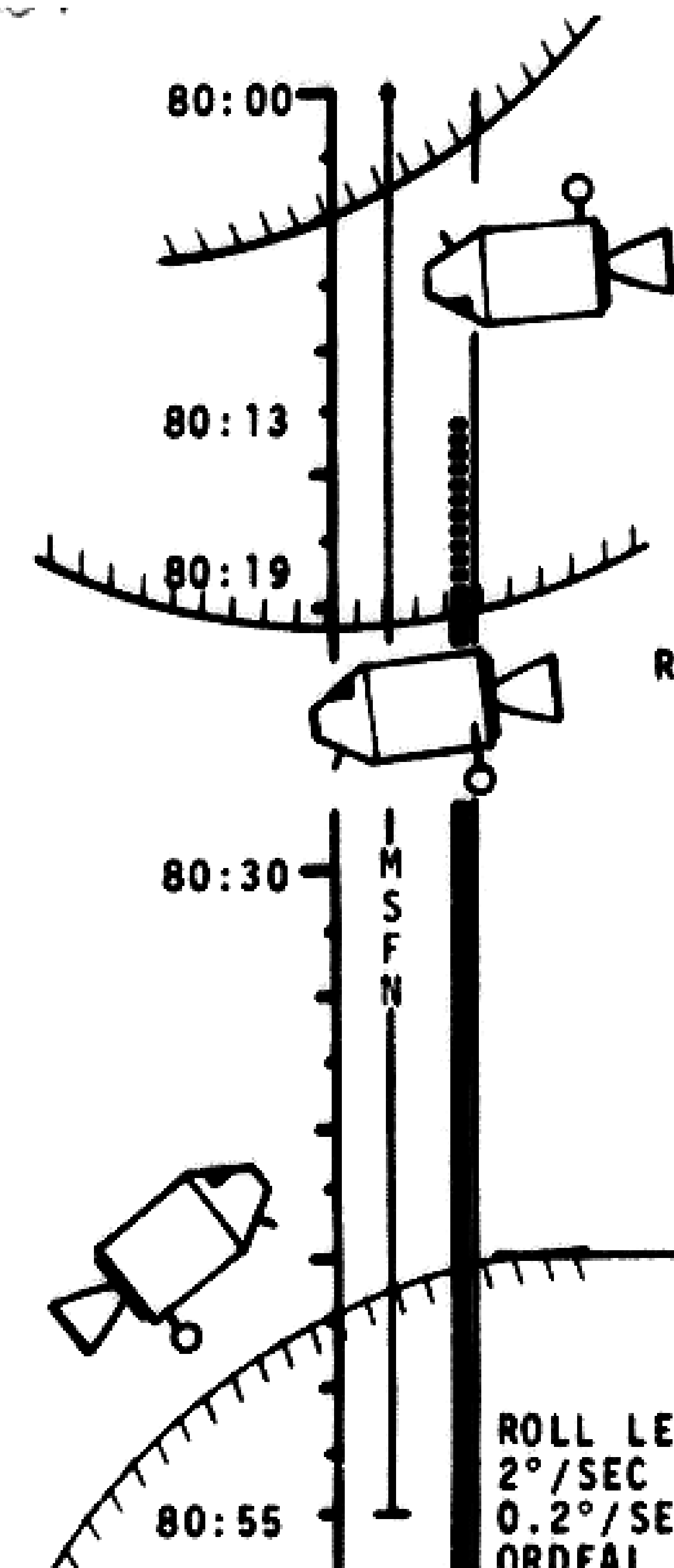
ACQUIRE MSFN ON OMNIS

REPORT NEW COORDINATES

MCC-H

ACQUIRE S/C

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
			79:00 - 80:00	4/LPO 5+6	2-61



80:00
80:13
80:19
80:30
80:55
81:00

M S F N

CDR

1000I

REST PERIOD
(3 HOURS)

ROLL LEFT 180° AT
2°/SEC PITCH UP AT
0.2°/SEC TO
ORDEAL 0,355, 0. BEGIN
ORB RATE

CMP

P22 AUTO OPTICS
IPI 80:09:08
TCA __:__:__
LDMK NO. B1
LAT +02.675°N
LONG/2 +17.512°
ALT -000.99 nm
NEW COORDINATES
LAT 02.457°
LONG/2 17.618°
ALT - 000.95 nm

V47 TRANS LM STATE
VECTOR TO CSM SLOT

IMU REALIGN P52
OPTION 3 - REFSMMAT
AND GYRO DRIFT TEST
STAR ID 20, 21
STAR ANGLE DIFF
00.01
TORQUE ANGLES:
X+ 00.052
Y+ 00.062
Z+ 00.071

16/SXT/~~NEW C XT~~ CEX 1/3 MAG
8/fps

LMP

PITCH DWN 0.3°/SEC AT ACQ

AT LOSS ROLL RIGHT 180°
AT 2°/SEC. GO INERTIAL

V64 ACQUIRE MSFN ON HGA

REPORT NEW COORDINATES

RECORD BLOCK DATA
(TEI_7) & MAP UPDATE

VERIFY SYSTEMS STATUS

1/80/B&W, 1/250 F 2.8 INT

2/150/B&W, 1/250 (R&B)

TRANS TO RIGHT COUCH

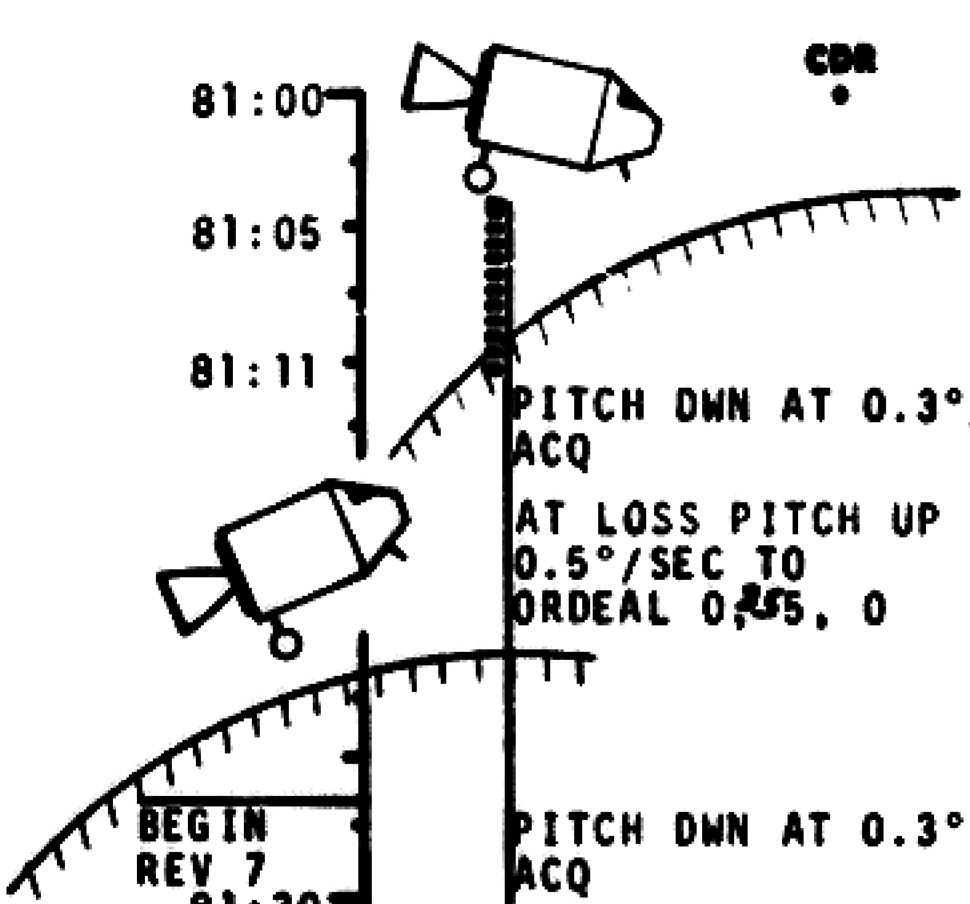
EAT PERIOD

MCC-H

ACQUIRE S/C
P27 UPDATE:
STATE
VECTOR
VOICE
UPDATE:
BLOCK DATA
& MAP
UPDATE 6/7
DUMP DSE

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	80:00 - 81:00	4/LPO 6	2-62

FLIGHT PLAN



81:00
81:05
81:11
BEGIN
REV 7
81:30
81:41
82:00

M S F N

CDR

PITCH DWN AT 0.3°/SEC AT
ACQ

AT LOSS PITCH UP AT
0.5°/SEC TO
ORDEAL 0,355, 0

PITCH DWN AT 0.3°/SEC AT
ACQ

AT LOSS PITCH UP AT
0.5°/SEC TO
ORDEAL 0,355, 0

PITCH DWN AT 0.3°/SEC AT
ACQ

AT LOSS PITCH UP AT
0.5°/SEC TO
ORDEAL 0,355, 0

CMP

CNTL PT SIGHTINGS (3)
START CAMERA AT IPI
ACQ OR TCA

P22 MAN ACQ SA 0° TA 10°
IPI __:__:__
ACQ __:__:__
CP1 NO. 504
LAT -05.250°S 1/60
LONG/2 -81.350°
ALT +000.00 nm

P22 AUTO OPTICS
IPI __:__:__
TCA __:__:__
CP2 NO. 526
LAT -10.200°S 1/500
LONG/2 +77.550°
ALT +000.00 nm

P22 MAN ACQ SA 0° TA 10°
IPI __:__:__
ACQ __:__:__
CP3 NO. 334
LAT -09.000°S 1/500
LONG/2 ~~+47.950°~~ +VR 1/75°
ALT +000.00 nm

PSEUDO LDG SITE SIGHTING
P22 AUTO OPTICS

LMP

BIOMED SW - LEFT

EAT PERIOD

NEW COORDINATES
LAT __.__°
LONG/2 __.__°
ALT __.__nm

EAT PERIOD

NEW COORDINATES
LAT __.__°
LONG/2 __.__°
ALT __.__nm

NEW COORDINATES
LAT __.__°
LONG/2 __.__°
ALT __.__nm

MCC-H

ACQUIRE S/C

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	81:00 - 82:00	4/LPO 6+7	2-63

FLIGHT PLAN

CDR

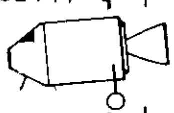
82:00

PITCH DWN AT 0.3°/SEC
AT ACQ

82:11

AT LOSS ROLL RIGHT 180°
AT 2°/SEC
GO INERTIAL

82:17



ACQUIRE MSFN ON HGA

REPORT NEW COORDINATES

RECORD MAP UPDATE &
BLOCK DATA (TEI_8)

M
S
F
N

82:30

GDC ALIGN TO IMU

ROLL LEFT 180° AT
2°/SEC TO HEADS UP

82:53

83:00

CMP

IPI | __ __:__ __:__ __
TCA | __ __:__ __:__ __
LDG SITE NO. B1
LAT +02.675°N
LONG/2 +17.512°
ALT -000.99 nm
NEW COORDINATES
LAT __ __.__ __ __°
LONG/2 __ __.__ __ __°
ALT __ __ __.__ __ nm

IMU REALIGN P52
OPTION 3 - REFSMMAT
AND GYRO DRIFT TEST
STAR ID __ __, __ __
STAR ANGLE DIFF __ __
__ __.__ __ __
TORQUE ANGLES:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __

16/SXT/CXT, ____, 6 fps

LMP

REST PERIOD
(2 HOURS)

MCC-H

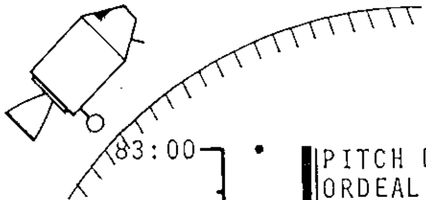
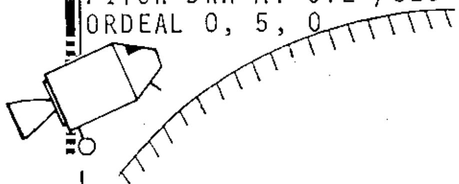
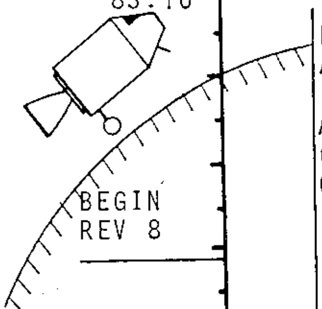
ACQUIRE S/C
P27 UPDATE:
STATE
VECTOR
VOICE
UPDATE:
MAP UPDATE
7/8
BLOCK DATA
DUMP DSE

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	82:00 - 83:00	4/LPO	2-64

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

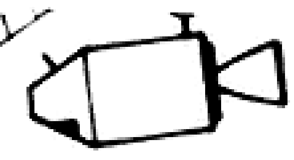




TIME	CDR	CMP	LMP	MCC-H
83:00	PITCH DWN AT 0.2°/SEC TO ORDEAL 0, 5, 0	CNTL PT SIGHTINGS (3)	REST PERIOD (2 HOURS)	
83:03		P22 AUTO OPTICS IPI __ __:__ __:__ __ TCA __ __:__ __:__ __ CP1 NO. 504 LAT -05.250°S LONG/2 -81.350° ALT +000.00 nm	NEW COORDINATES LAT __ __.__ __ __° LONG/2 __ __.__ __ __° ALT __ __ __.__ __nm	
83:10	PITCH DWN AT 0.3°/SEC AT ACQ AT LOSS PITCH UP AT 0.5°/SEC TO ORDEAL 0, 5, 0	P22 AUTO OPTICS IPI __ __:__ __:__ __ TCA __ __:__ __:__ __ CP2 NO. 526 LAT -10.200°S LONG/2 +77.550° ALT +000.00 nm	NEW COORDINATES LAT __ __.__ __ __° LONG/2 __ __.__ __ __° ALT __ __ __.__ __nm	
BEGIN REV 8				
83:30	PITCH DWN AT 0.3°/SEC AT ACQ PITCH UP AT 0.5°/SEC TO ORDEAL 0, 5, 0			
83:39	ACQUIRE MSFN ON OMNIS	P22 AUTO OPTICS IPI __ __:__ __:__ __ TCA __ __:__ __:__ __ CP3 NO. 334 LAT -09.100°S LONG/2 +47.950° ALT +000.00 nm	NEW COORDINATES LAT __ __.__ __ __° LONG/2 __ __.__ __ __° ALT __ __ __.__ __nm	ACQUIRE S/C
MSFN	PITCH DWN AT 0.3°/SEC AT ACQ PITCH UP AT 0.5°/SEC TO ORDEAL 0, 5, 0	PSEUDO LDG SITE SIGHTING P22 AUTO OPTICS		
84:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	83:00 - 84:00	4/LPO	2-65

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH



84:00

84:10

84:16

84:30

84:51

85:00

M S F N

CDR

PITCH DWN AT 0.3°/SEC
AT ACQ

AT LOSS ROLL RIGHT 180°
AT 2°/SEC (HEADS DWN)
GO INERTIAL

ACQUIRE MSFN ON HGA

GDC ALIGN

PITCH UP AT 0.2°/SEC TO
ORDEAL 180, 0, 0 (HEADS
DWN) BEGIN ORB RATE

16/18/NEW CEX, 1/250, ~~f2.8~~, 1 fps F4

CMP

IPI __ __:__ __:__ __
TCA __ __:__ __:__ __
LDMK NO. B1
LAT +02.512°N 1/60
LONG/2 +17.512°
ALT +000.99 nm

LiOH CANISTER CHANGE
(CARTRIDGE 10 FROM B~~5~~6
INTO CANISTER B)
V47 TRANS LM STATE
VECTOR TO CSM SLOT

IMU REALIGN P52
OPTION 3 - REFSMMAT
AND GYRO DRIFT TEST
STAR ID __ __, __ __
STAR ANGLE DIFF __

TORQUE ANGLES:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __

REST PERIOD
(2 HOURS)

LMP

NEW COORDINATES
LAT __ __.__ __ __°
LONG/2 __ __.__ __ __°
ALT __ __ __.__ __ nm

REST PERIOD
(2 HOURS)

2/150/~~NEW~~ B&W, 1/250
(R&B)

BIOMED SW - RIGHT

VERIFY SYSTEMS STATUS

RECORD MAP UPDATE &
BLOCK DATA (TEI_9)

1/80/H.S. B, f2.8, POL
POSIGRADE HORIZON

SR -12 MIN: START ~~DARK~~ ZODIACAL
3 1 MIN EXP, Δ POL 60°

1/80/~~B&W, 1/4, f4~~, POL
HS, B, F2.8

SR -4 MIN: START CORONA
3 10SEC EXP, Δ POL 60°
SR-3 MIN: 3 4SEC EXP, Δ POL 60°

MCC

ACQUIRE S/C

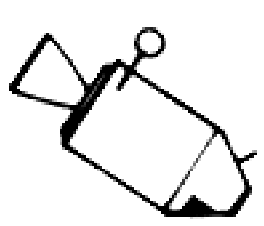
P27 UPDATE:
STATE
VECTOR

DUMP DSE

VOICE
UPDATE:
MAP UPDATE
8/9
BLOCK DATA

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	84:00 - 85:00	4/LPO 8	2-66

SR-1 MIN: SEVERAL 1/60 SEC EXP, Δ POL 60°
SR-15SEC: 1/60 SEC EXP TO SUNRISE, Δ POL 60° AS FEASIBLE



FLIGHT PLAN

85:00

85:02

85:08

BEGIN
REV 9

85:30

85:37

86:00

M S F N

CDR

AT S/C SR
PITCH UP 60° AT 0.5°/SEC
TO ORDEAL 180, 300, 0
CONTINUE ORB RATE
START CAMERA AT
S/C SR +12 MIN

ZERO PHASE TARGETS OF
OPPORTUNITY

PITCH UP 40° AT 0.2°/SEC
TO ORDEAL 180, 260, 0

ROLL RT AS REQD TO SHADE
WINDOWS-KEEP X-AXIS +10°
ON TRACK

APPROACH TERM AT ORDEAL
0, 260, 0 ORB RATE

CMP

REST PERIOD
(2 HOURS)

LMP

1/80/NEW C BRKT INT
1/250 F4

START CAMERA 1 AT
SR +12 MIN

CONVERGENT STEREO
PHOTOGRAPHY-EXTRA
EXP EACH 5 MIN -
NOTE TIME

ACQUIRE MSFN ON HGA

RECORD MAP UPDATE

2/80 B&W/R&B
4 STOPS WIDER

MCC-H

ACQUIRE S/C

VOICE
UPDATE:
MAP UPDATE
9/10

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
			85:00 - 86:00	4/LPO 8-9	2-67

CDR

86:00

86:08

STOP CAMERA AT TERM

86:14

ROLL RIGHT 180° IF REQD
PITCH DWN 60° AT
0.2°/SEC TO
ORDEAL 180, 310, 0
(HEADS DWN)
GO INERTIAL

V47 TRANS LM STATE VECTOR TO CSM SLOT

86:30

MSFN

86:50

PITCH DWN AT 0.2°/SEC TO
ORDEAL 180, ~~180~~ 270, 0
BEGIN ORB RATE
(HEADS DWN +Z FWD)

87:00

CMP

REST PERIOD
(2 HOURS)

IMU REALIGN P52
OPTION 3 - REFSMMAT
AND GYRO DRIFT TEST
STAR ID ___, ___
STAR ANGLE DIFF ___

TORQUE ANGLES:
X ___ ___ . ___ ___ ___
Y ___ ___ . ___ ___ ___
Z ___ ___ . ___ ___ ___

LMP

STOP CAMERA AT TERM
2/80/HS, 1/4 f2.8

RECORD BLOCK DATA
(TEI_{10}) & MNVR PAD

VERIFY SYSTEMS STATUS

1/80/NEW B&W 1/250,
f2.8 INT (CDR)
2/150/C, 1/250

MCC-H

P27 UPDATE:
STATE
VECTOR

VOICE
UPDATE:
BLOCK DATA
MNVR PAD

DUMP DSE

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	86:00 - 87:00	4/LPO 9	2-68

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

CDR

87:00
87:01

87:07

MNVR TO TEI ATT

BEGIN
REV 10

PITCH UP AT 0.2°/SEC TO
ORDEAL 180, ~~180~~ 270, 0
BEGIN ORB RATE

EAT PERIOD

87:30

87:36

MSFN

88:00

CMP

SXT STAR CK

EAT PERIOD

LMP

EAT PERIOD

2/150/B&W OR C,1/250
ACQUIRE MSFN ON HGA

IMAGE MOTION
16/200/CEX SPOT
1/250 (1/3 MAG)

RECORD MNVR PAD, BLOCK
DATA (TEI_{11}), MAP UPDATE

MCC-H

ACQUIRE S/C

DUMP DSE
PIPA BIAS CK
P27 UPDATE:
STATE
VECTOR *(CSM+LM)*
TARGET LOAD

VOICE
UPDATE:
MNVR PAD
BLOCK DATA
MAP UPDATE
10

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
		... 1968	87:00 - 88:00	4/LPO ~~9~~ 10	2-69

88:00

88:07

88:13

88:30

88:48

89:00

MSFN

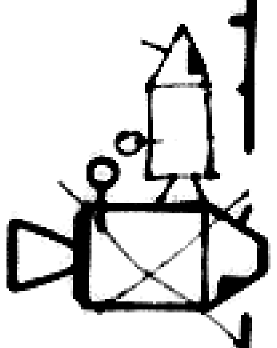
GO/NO GO FOR TEI

GO INERTIAL

EXTERNAL ΔV P30
EMS TEST
SPS THRUST P40

MNVR TO BURN ATT

PUGS MODE Sw - AUX



IMU REALIGN P52
OPTION 3 - REFSMMAT
AND GYRO DRIFT TEST
STAR ID
STAR ANGLE DIFF __ __

TORQUE ANGLES: __ __
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __

SXT STAR CK
TRANSFER TO COUCH

2/150/C SPOT
16/75/CEX/1 fps/SPOT

PRE TEI SYSTEMS CKS
C&W CK
CM RCS CK
SM RCS CK
SPS PERIODIC MONITOR
EPS PERIODIC MONITOR
ECS PERIODIC MONITOR

BIOMED Sw - LEFT

GO/NO GO

BURN STATUS REPORT

X X	:	ΔTIG
X X	:	BT
	.	V_{gx}

——TRIM——

X X X		R
X X X		P
X X X		Y
☐	.	V_{gx}
☐	.	V_{gy}
☐	.	V_{gz}
☐	.	ΔV_c
X X X		FUEL
X X X		OX
X X X		UNBALANCE

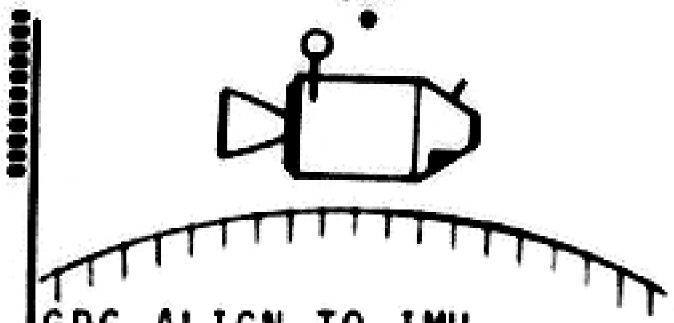
REMARKS:

TEI
BURN CHART

	P or Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
TEI	10°/SEC TAKEOVER	10° TAKEOVER	B/T+2 SEC & ΔV_c=-40 fps	TRIM TO 2.0 fps

TEI ABORT MODES-SYSTEMS PROBLEMS:15-MIN ABORT CHART OTHERWISE

TEI V_{go}	B/T	TRAJECTORY	ABORT MODE
2850-950	0 -2:00	LUNAR ORBIT	MODE III AFTER 1 REV
950-600	2:00-2:20	UNSTABLE	5-HR COAST, MODE I
600-0	2:20-2:54	HYPERBOLIC	COAST OUT OF SPHERE - P37

Time	CDR	CMP	LMP	MCC-H
89:00	 GDC ALIGN TO IMU	SM RCS MON CK GETI = 89:15:07 ΔV = 3532 fps BT = 3 MIN 26 SEC		
89:15	TEI PITCH UP TO VERTICAL (+Z FWD) TO ACQ MOON IN WINDOW CDR CREW STATUS REPORT	SM RCS MON CK V 66 E TRANS CSM STATE VECTOR TO LM SLOT	SPS MON CK ACQUIRE MSFN WITH HGA TEI BURN STATUS REPORT INITIATE BAT A CHARGE	ACQUIRE S/C
89:30	SLEEP PERIOD (5 HOURS)	REINITIALIZE W MATRIX R_1 +00094 R_2 ~~+00571~~ 00057 R_3 +00003	BIOMED Sw - CENTER *TERMINATE WATER BOILER EVAP* *1/60 (250)/B+W; RAB SPOT, 4 STOPS WIDER*	
89:45			*SC EXT ATM (POSTBURN) 1/80/HS F2.8 B 6SEC, B 1SEC, 1/60 90° FROM SUN*	DUMP DSE UPDATE ENTRY REFSMMAT
90:00				

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	89:00 - 90:00	4/TEC	2-71

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

Time	CDR	CMP	LMP	MCC-H
90:00		IMU REALIGN P52 OPTION 1 - PREFERRED STAR ID __ __ STAR ANGLE DIFF __ __ TORQUE ANGLES: __ __ X __ __ . __ __ __ Y __ __ . __ __ __ Z __ __ . __ __ __	MNVR TO P52 ATT	
90:15				
90:30	SLEEP PERIOD (5 HOURS)	TRN BIAS CISLUNAR NAVIGATION P23 1. STAR 01 LNH *00 210* STAR__ __ L__H 2 SETS	MNVR TO SIGHTING ATT	
90:45		2. STAR 02 LNH *00 210* STAR__ __ L__H		
91:00				

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	90:00 - 91:00	4/TEC	2-72

Time		CDR	CMP	LMP	MCC-H
91:00	MSFN	SLEEP PERIOD (5 HOURS)			
91:15			TRN BIAS 3. STAR 10 LFH 00220 STAR__ __ L__H 2 SETS	MNVR TO SIGHTING ATT	
91:30					
91:45			4. STAR 11 LFH 00220 STAR__ __ L__H 2 SETS		
92:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	91:00 - 92:00	4/TEC	2-73

FLIGHT PLAN

Time		CDR	CMP	LMP	MCC-H
92:00	MSFN	SLEEP PERIOD (5 HOURS)		MNVR TO PTC ATT P 029 Y 315 P____ Y____ ROLL 0.1°/SEC	
92:15					
92:30					
92:45					
93:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	92:00 - 93:00	4/TEC	2-74

	CDR	CMP	LMP	MCC-H
93:00				
93:30	SLEEP PERIOD (5 HOURS)	EAT PERIOD	EAT PERIOD	
			BIOMED SW - LEFT CMP/LMP CREW STATUS REPORT	
94:00				
94:30	EAT PERIOD	SLEEP PERIOD (5 HOURS)	SLEEP PERIOD (~~7~~ 5 HOURS)	
95:00				

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	93:00 - 95:00	4/TEC	2-75

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
95:00	CDR CREW STATUS REPORT CHLORINATE WATER			
95:15				
95:30		SLEEP PERIOD (5 HOURS)	SLEEP PERIOD (5 HOURS)	
95:45	LiOH CANISTER CHANGE (CARTRIDGE NO. 11 FROM ~~B5~~ A2 INTO CANISTER ~~B~~ A)			
96:00	MNVR TO P52 ATT			

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
				4/TEC	2-76

96:00	CDR	CMP	LMP	MCC-H
96:00	IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID ___ ___ STAR ANGLE DIFF ___ ___ TORQUE ANGLES: X ___ ___ . ___ ___ ___ Y ___ ___ . ___ ___ ___ Z ___ ___ . ___ ___ ___	SLEEP PERIOD (5 HOURS)	SLEEP PERIOD (5 HOURS)	
96:15				
96:30	MNVR TO PTC ATT P 029 Y 315 P ___ Y ___ ROLL 0.1°/SEC			
96:45				
97:00				

M S F N

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	96:00 - 97:00	5/TEC	2-77

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

72:00 MSFN
P52
Rev 3 LOI2
TRAINING PHOTOS
MSFN
P52
Rev 4 VERTICAL STERO
MSFN
LMK LIGHTING EVAL
P52
Rev 5 LMK SIGHTING (1)
78:00 LDG SITE SIGHTING
MSFN
P52
Rev 6 LMK SIGHTING (1)
LDG SITE SIGHTING
MSFN
P52
Rev 7 LMK SIGHTINGS (3)
LDG SITE SIGHTING
MSFN
P52
Rev 8 LMK SIGHTINGS (3)
84:00

CDR EAT PERIOD

LMP EAT PERIOD

CMP REST PERIOD (2 HOURS)

CDR REST PERIOD (3 HOURS)

LMP REST PERIOD (2 HOURS)

84:00 LDG SITE SIGHTING
MSFN
P52
DARKSIDE PHOTOGRAPHY
Rev 9 CONVERGENT STERO AND TV
MSFN
P52
Rev 10
MSFN
P52
89:15 TEI
90:00 P52
CMP - STAR/LUNAR HORIZON SIGHTINGS
PTC
96:00

EAT PERIOD (ALL)

CMP & LMP EAT PERIOD

CDR EAT PERIOD

CMP REST PERIOD (2 HOURS)

CDR SLEEP PERIOD (5 HOURS)

CMP & LMP SLEEP PERIOD (5 HOURS)

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	72:00 - 96:00	4	5-4

FLIGHT PLANNING BRANCH

FLIGHT PLAN

		CDR	CMP	LMP	MCC-H
97:00					
97:30					
98:00	MSFN		SLEEP PERIOD (5 HOURS)	SLEEP PERIOD (5 HOURS)	
98:30					
99:00					

Time	MSFN	CDR	CMP	LMP
99:00				BIOMED SW - CENTER
99:30	MSFN	EAT PERIOD	EAT PERIOD	EAT PERIOD

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	99:00 - 100:00	5/TEC	2-79

FLIGHT PLAN

Time	MSFN	CDR	CMP	LMP	MCC-H
100:00		MNVR TO P52 ATT	IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID ___ , ___ ___ STAR ANGLE DIFF ___ . ___ TORQUE ANGLES: X ___ ___ . ___ ___ ___ Y ___ ___ . ___ ___ ___ Z ___ ___ . ___ ___ ___	CMP/LMP CREW STATUS REPORT	
100:15		MNVR TO SIGHTING ATT	TRN BIAS		
100:30	MSFN		CISLUNAR NAVIGATION P23 1. STAR 02 LNH 00210 STAR ___ ___ L ___ H 1 SET 2. STAR 11 LFH 00220 STAR ___ ___ L ___ H 1 SET		
100:45			3. STAR 01 LNH 00210 STAR ___ ___ L ___ H 1 SET		
101:00		MNVR TO SIGHTING ATT			

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	100:00 - 101:00	5/TEC	2-80

101:00

101:15

TRN BIAS

CISLUNAR NAVIGATION P23

1. STAR 22 EFH *00120*
   STAR__ __ E__H
   2 SETS

101:30

M
S
F
N

2. STAR 26 ENH *00110*
   STAR__ __ E__H
   2 SETS

101:45

BIOMED Sw - RIGHT

102:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	101:00 - 102:00	5/TEC	2-81

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FLIGHT PLANNING BRANCH

FLIGHT PLAN

CDR | CMP | LMP | MCC-H

102:00

3. STAR 31 ENH *00110*
   STAR__ __ E__H
   2 SETS

102:15

P27 UPDATE
STATE
VECTOR
TGT LOAD
REFSMMAT

102:30

M
S
F
N

VOICE
UPDATE:
MNVR PAD

MNVR TO P52 ATT

RECORD MNVR PAD

IMU REALIGN P52
OPTION 3 - RESFMMAT
STAR ID __ __, __ __
STAR ANGLE DIFF __.__ __

102:45

TORQUE ANGLES:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __

103:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
			102:00 103:00	5/TEC	2-82

BURN STATUS REPORT

X X ☐	:		ΔTIG
X X	:		BT
☐		.	V_{gx}

TRIM

X X X			R
X X X			P
X X X			Y
☐		.	V_{gx}
☐		.	V_{gy}
☐		.	V_{gz}
☐		.	ΔV_c
X X X			FUEL
X X X			OX
X X X			UNBALANCE

REMARKS:

2-82a

MCC'S

BURN CHART

	P or Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
MCC(ALL)	10°/SEC TAKEOVER	10° TAKEOVER	B/T +1 SEC	TRIM TO 0.2 fps

FLIGHT PLAN

TIME		CDR	CMP	LMP	MCC-H
103:00					
103:15		V47 TRANS LM STATE VECTOR TO CSM SLOT →			
		EXT ΔV P30			
	MSFN	SPS/RCS THRUST P40/41			
103:30		MNVR TO BURN ATT	SXT STAR CK		
103:45		EMS TEST	TRANS TO COUCH		PIPA BIAS CK
			SM RCS MON CK		
TEI+15		GDC ALIGN TO IMU			
104:00		MCC_5 ΔV=NOMINALLY ZERO			

	CDR	CMP	LMP	MCC-H
104:00	V66 TRANS CSM STATE VECTOR TO LM SLOT	SM RCS MON CK	MCC_5 BURN STATUS REPORT ~~INITIATE BAT CHARGE~~	
	UNSTOW TV CAMERA			
104:15	TV			
104:30	EAT PERIOD	EAT PERIOD	EAT PERIOD	
	STOW TV CAMERA			
104:45				
105:00	CDR CREW STATUS REPORT			

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	104:00 - 105:00	5/TEC	2-84

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FLIGHT PLANNING BRANCH

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
105:00		LiOH CANISTER CHANGE (CARTRIDGE NO. 12 FROM ~~B6~~ A2 INTO CANISTER B)	BIOMED Sw - CENTER TRANS TO LH COUCH MNVR TO SIGHTING ATT	
105:15		TRN BIAS CISLUNAR NAVIGATION P23 1. STAR 22 EFH 00120 STAR__ __ E__H 2 SETS		
105:30	SLEEP PERIOD (7 HOURS)			
		2. STAR 26 ENH 00110 STAR__ __ E__H 1 SET		
105:45				
		3. STAR 31 ENH 00110 STAR__ __ E__H 2 SETS		
106:00				

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

	CDR	CMP	LMP	MCC-H
106:00 – 107:00 MSFN	SLEEP PERIOD (7 HOURS)	TRN BIAS 1. STAR 02 LNH 00210 STAR__ __ L__H 1 SET 2. STAR 11 LFH 00220 STAR__ __ L__H 1 SET 3. STAR 01 LNH 00210 STAR__ __ L__H 1 SET RETURN TO EARTH P37	MNVR TO SIGHTING ATT	

Time marks: 106:00, 106:15, 106:30, 106:45, 107:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	106:00 - 107:00	5/TEC	2-86

MSC Form 1910 (Nov 68) FLIGHT PLANNING BRANCH

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
107:00 – 108:00 MSFN	SLEEP PERIOD (7 HOURS)	IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID __ __, __ __ STAR ANGLE DIFF __ __ . __ __ TORQUE ANGLES: X __ __ . __ __ __ Y __ __ . __ __ __ Z __ __ . __ __ __ TRN BIAS	MNVR TO P52 ATTITUDE MNVR TO SIGHTING ATT	

Time marks: 107:00, 107:15, 107:30, 107:45, 108:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
[illegible]	FINAL	November 22, 1968	107:00 - 108:00	5/TEC	2-87

108:00 CDR CMP LMP MCC-H

TIME	CDR	CMP	LMP	MCC-H
108:00	SLEEP PERIOD (7 HOURS)	CISLUNAR NAVIGATION P23 1. STAR 22 EFH 00120 STAR__ __ E__H 1 SET	BIOMED Sw - RIGHT	
108:15				
108:30		2. STAR 26 ENH 00110 STAR__ __ E__H 2 SETS		
108:45		3. STAR 31 ENH 00110 STAR__ __ E__H 2 SETS	MNVR TO PTC ATT P 029 Y 315 P__ Y__ ROLL 0.1°/SEC P&Y FREE REESTABLISH PTC AT +15° IN P OR Y AND RECORD GET	
109:00		RETURN TO EARTH P37		

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	108:00 - 109:00	5/TEC	2-88

FLIGHT PLAN

TIME	CDR	CMP	LMP	MCC-H
109:00	SLEEP PERIOD (7 HOURS)			
109:30				
110:00				
110:30				
111:00				

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	109:00 - 111:00	5/TEC	2-89

	MSFN	CDR	CMP	LMP	MCC-H
111:00					
111:30		SLEEP PERIOD (7 HOURS)	EAT PERIOD	EAT PERIOD	
				CMP/LMP CREW STATUS REPORT	
				BIOMED Sw - LEFT	
112:00					
112:30		EAT PERIOD	SLEEP PERIOD (7 HOURS)	SLEEP PERIOD (7 HOURS)	
113:00					

FLIGHT PLAN

	MSFN	CDR	CMP	LMP	MCC-H
113:00					
		CDR CREW STATUS REPORT			
113:30					
114:00			SLEEP PERIOD (7 HOURS)	SLEEP PERIOD (7 HOURS)	
114:30					
		MNVR TO P52 ATT			
115:00					

	CDR	CMP		LMP	MCC-H
115:00	IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID ___ STAR ANGLE DIFF ___.___ TORQUE ANGLES: X ___.___ Y ___.___ Z ___.___				
115:15	MNVR TO PTC ATT P 029 Y 315 P ___ Y ___ ROLL 0.1°/SEC				
115:30 MSFN		SLEEP PERIOD (7 HOURS)		SLEEP PERIOD (7 HOURS)	
115:45					
116:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	115:00 - 116:00	5/TEC	2-92

FLIGHT PLAN

	CDR	CMP		LMP	MCC-H
116:00					
116:30					
117:00 MSFN		SLEEP PERIOD (7 HOURS)		SLEEP PERIOD (7 HOURS)	
117:30					
118:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	116:00 - 118:00	5/TEC	2-93

CDR | CMP | LMP

118:00
118:30
119:00
119:30
120:00

M S F N

CDR: EAT PERIOD

LiOH CANISTER CHANGE (CARTRIDGE NO. 13 FROM ~~B6~~ A2 INTO CANISTER A)

MNVR TO SIGHTING ATT

CMP: SLEEP PERIOD (7 HOURS)

EAT PERIOD

TRN BIAS

LMP: SLEEP PERIOD (7 HOURS)

BIOMED Sw - CENTER

EAT PERIOD

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	118:00 - 120:00	5/TEC	2-94

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FLIGHT PLANNING BRANCH

FLIGHT PLAN

96:00

P52
PTC

CMP & LMP SLEEP PERIOD (5 HOURS)

EAT PERIOD ALL

P52

CMP - STAR/LUNAR HORIZON SIGHTINGS

102:00

CMP - STAR/EARTH HORIZON SIGHTINGS

P52

TEI+15 MCC5

TV

EAT PERIOD ALL

CMP - STAR/EARTH HORIZON SIGHTINGS

CMP - STAR/LUNAR HORIZON SIGHTINGS

CDR SLEEP PERIOD (7 HOURS)

P52

108:00

108:00

CMP - STAR/ EARTH HORIZON SIGHTINGS

PTC (P & Y FREE)

CDR SLEEP PERIOD (7 HOURS)

CMP & LMP EAT PERIOD

CDR EAT PERIOD

CMP & LMP SLEEP PERIOD (7 HOURS)

114:00

P52

CDR EAT PERIOD

CMP & LMP EAT PERIOD

120:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
			96:00 - 120:00	5	5-5

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
120:00		CISLUNAR NAVIGATION P23 1. STAR 22 EFH 00120 STAR__ __ E__H 1 SET 2. STAR 26 ENH 00110 STAR__ __ E__H 1 SET	CMP/LMP CREW STATUS REPORT	
120:15				
120:30 (MSFN)		3. STAR 31 ENH 00110 STAR__ __ E__H 1 SET RETURN TO EARTH P37 CHLORINATE WATER		
120:45			RECORD MNVR PAD (INITIATE BAT B CHARGE (FOLLOWED BY BAT A)	P27 UPDATE: STATE VECTOR TGT LOAD VOICE UPDATE: MNVR PAD
121:00				

BURN STATUS REPORT

X X [ ] :	ΔTIG
X X :	BT
[ ] .	V_{gx}
TRIM	
X X X	R
X X X	P
X X X	Y
[ ] .	V_{gx}
[ ] .	V_{gy}
[ ] .	V_{gz}
[ ] .	ΔV_c
X X X	FUEL
X X X	OX
X X X	UNBALANCE

REMARKS:

MCC'S

BURN CHART

	P or Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
MCC(ALL)	10°/SEC TAKEOVER	10° TAKEOVER	B/T +1 SEC	TRIM TO 0.2 fps

FLIGHT PLAN

TIME		CDR	CMP	LMP	MCC-H
121:00		MNVR TO P52 ATT			
			IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID ___ . ___ STAR ANGLE DIFF ___ . ___ TORQUE ANGLES: X ___ ___ . ___ ___ ___ Y ___ ___ . ___ ___ ___ Z ___ ___ . ___ ___ ___		
121:15					
		V47 TRANS LM STATE VECTOR TO CSM SLOT			
	MSFN	EXT ΔV P30			
121:30		SPS/RCS THRUST P40/41 MNVR TO BURN ATT			
			SXT STAR CK		
121:45		EMS TEST	TRANS TO COUCH		PIPA BIAS CK
		GDC ALIGN	SM RCS MON CK		
TEI + 33 HRS		MCC_6 ΔV=NOMINALLY ZERO			
122:00					

TIME		CDR	CMP	LMP	
122:00		V66 TRANS CSM STATE VECTOR TO LM SLOT	SM RCS MON CK	MCC$_6$ BURN STATUS REPORT ~~INITIATE BAT CHARGE~~	
				BIOMED Sw - RIGHT	
122:15					
122:30	MSFN	MNVR TO SIGHTING ATT			
			TRN BIAS CISLUNAR NAVIGATION P23 1. STAR 02 LNH 00210 STAR__ __ L__H 2 SETS		
122:45					
			2. STAR 01 LNH 00210 STAR__ __ L__H 1 SET		
123:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	122:00 - 123:00	6/TEC	2-97

FLIGHT PLAN

TIME		CDR	CMP	LMP	MCC-H
123:00			RETURN TO EARTH P37		
		MNVR TO SIGHTING ATT			
123:30					
123:30	MSFN				
			TRN BIAS CISLUNAR NAVIGATION P23 1. STAR 22 EFH 00120 STAR__ __ E__H 2 SETS		
123:45					
			2. STAR 26 ENH 00110 STAR__ __ E__H 1 SET		
124:00					

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
			[illegible] 124:00	6/TEC	2-98

CDR | CMP

124:00

3. STAR 31 ENH 00110
STAR__ __ E__H
2 SETS

124:15

124:30

MSFN

MNVR TO PTC ATT
P 029 Y 315
P ___ Y ___
ROLL 0.1°/SEC

RETURN TO EARTH P37

124:45

125:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	124:00 - 125:00	6/TEC	2-99

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FLIGHT PLANNING BRANCH

FLIGHT PLAN

CDR | CMP | LMP | MCC-H

125:00

BIOMED Sw - CENTER

125:30

MSFN

126:00

126:30

EAT PERIOD

EAT PERIOD

EAT PERIOD

CDR CREW STATUS REPORT

127:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
		November 22,1968	126:00 - 127:00	6/TEC	2-100

	CDR	CMP	LMP	MCC-H
127:00			LiOH CANISTER CHANGE (CARTRIDGE NO. 14 FROM ~~B6~~ A2 INTO CANISTER B)	
127:30				
			BIOMED Sw - RIGHT	
			UNSTOW TV CAMERA	
128:00 MSFN	SLEEP PERIOD (7 HOURS)		TV	
			STOW TV CAMERA	
128:30				
129:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	127:00 - 129:00	6/TEC	2-101

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
129:00				
129:15				
129:30 MSFN	SLEEP PERIOD (7 HOURS)		MNVR TO P52 ATT	
		IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID __ , __ STAR ANGLE DIFF __.__ __ TORQUE ANGLES: X __ __.__ __ __ Y __ __.__ __ __ Z __ __.__ __ __		
129:45				
		TRN BIAS	MNVR TO SIGHTING ATT	
130:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	129:00 - 130:00	6/TEC	2-102

	CDR	CMP	LMP	MCC-H
130:00		CISLUNAR NAVIGATION P23		
		1. STAR 02 LNH 00210 STAR__ __ L__H 2 SETS		
130:15				
130:30	M S F N SLEEP PERIOD (7 HOURS)		MNVR TO SIGHTING ATT	
130:45		TRN BIAS		
		1. STAR 22 EFH 00120 STAR__ __ E__H 2 SETS		
131:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	130:00 - 131:00	6/TEC	2-103

MSC Form 1910 (Nov 68) FLIGHT PLANNING BRANCH

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
131:00			BIOMED Sw - CENTER	
		2. STAR 26 ENH 00110 STAR__ __ E__H 2 SETS		
131:15				
131:30	M S F N SLEEP PERIOD (7 HOURS)	3. STAR 31 ENH 00110 STAR__ __ E__H 2 SETS		
131:45				
		~~RETURN TO EARTH P37~~	MNVR TO PTC ATT P 029 Y 315 P Y ROLL 0.1°/SEC	
132:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
		1968	131:00 - 132:00	6/TEC	2-104

	CDR	CMP	LMP	MCC-H
132:00 – 134:00 (MSFN)	SLEEP PERIOD (7 HOURS)	RETURN TO EARTH P37 (133:00); EAT PERIOD	STATE VECTOR GO/NO GO; RECORD BLOCK DATA FOR EI -2 HOURS + ENTRY PAD; EAT PERIOD (133:00); CMP/LMP CREW STATUS REPORT; BIOMED Sw - LEFT	EI -15 HRS P27 UPDATE: STATE VECTOR GO/NO GO; VOICE UPDATE: BLOCK DATA FOR EI -2 HOURS MCC + ENTRY PAD

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	132:00 - 134:00	6/TEC	2-105

MSC Form 1910 (Nov 68) FLIGHT PLANNING BRANCH

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
134:00 – 135:00 (MSFN)	EAT PERIOD	SLEEP PERIOD (7 HOURS)	SLEEP PERIOD (7 HOURS)	

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
			134:00 - 135:00	6/TEC	2-106

	CDR	CMP	LMP	MCC-H
135:00	BEGIN CABIN COLD SOAK CDR CREW STATUS REPORT			
135:15				
MSFN	MNVR TO P52 ATT IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID __ __, __ __ STAR ANGLE DIFF __ . __ __ TORQUE ANGLES: X __ __ . __ __ __ Y __ __ . __ __ __ Z __ __ . __ __ __	SLEEP PERIOD (7 HOURS)	SLEEP PERIOD (7 HOURS)	
135:30				
135:45				
	REESTABLISH PTC P 029 Y 315 P __ Y __ ROLL 0.1°/SEC			
136:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	135:00 - 136:00	6/TEC	2-107

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
136:00				
136:30				
MSFN		SLEEP PERIOD (7 HOURS)	SLEEP PERIOD (7 HOURS)	
137:00				
137:30				
138:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
	[illegible]	November 22, 1968	136:00 - 138:00	6/TEC	2-108

CDR | CMP | LMP | MCC-H

138:00

138:15

MSFN

CMP: SLEEP PERIOD (7 HOURS)

LMP: SLEEP PERIOD (7 HOURS)

138:30

138:45

139:00

FLIGHT PLAN

CDR | CMP | LMP | MCC-H

139:00

LiOH CANISTER CHANGE
(CARTRIDGE NO. 15 FROM
~~A2~~ INTO CANISTER A)
#3

139:15

MNVR TO P52 ATT

IMU REALIGN P52
OPTION 3 - REFSMMAT
STAR ID ___ ___
STAR ANGLE DIFF ___.___ ___

139:30

TORQUE ANGLES:
X ___ ___.___ ___ ___
Y ___ ___.___ ___ ___
Z ___ ___.___ ___ ___

MSFN

CMP: SLEEP PERIOD (7 HOURS)

LMP: SLEEP PERIOD (7 HOURS)

REESTABLISH PTC
P 029 Y 315
P ___ Y ___
ROLL 0.1°/SEC

139:45

140:00

	CDR	CMP	LMP	MCC-H
140:00 – 141:00 (MSFN)	EAT PERIOD	SLEEP PERIOD (7 HOURS)	SLEEP PERIOD (7 HOURS)	

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	140:00 - 141:00		2-111

FLIGHT PLAN

	CDR	CMP	LMP	MCC-H
141:00 (MSFN)	MNVR TO SIGHTING ATT	LOSS OF COMM CONTINGENCY TRN BIAS CISLUNAR NAVIGATION P23 1. STAR 22 EFH 00120 STAR__ __ E__H 1 SET 2. STAR 26 ENH 00110 STAR__ __ E__H 1 SET 3. STAR 31 ENH 00110 STAR__ __ E__H 1 SET	BIOMED Sw - RIGHT	
141:30 – 142:00		EAT PERIOD	EAT PERIOD	

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
		November 22,1968	141:00 - 142:00		2-112

142:00
CDR
EAT PERIOD
EAT PERIOD
142:15
142:30
M S F N
LOSS OF COMM CONTINGENCY
TRN BIAS
CISLUNAR NAVIGATION P23
1. STAR 22 EFH 00120
STAR__ __ E__H
1 SET
2. STAR 26 ENH 00110
STAR__ __ E__H
1 SET
142:45
143:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22,1968	142:00 - 143:00		2-113

FLIGHT PLAN

120:00
CMP - STAR/EARTH HORIZON SIGHTINGS 3 SETS
P52
TEI+33
MCC6
CMP - STAR/LUNAR HORIZON SIGHTINGS
CMP - STAR/EARTH HORIZON SIGHTINGS
PTC
126:00
EAT PERIOD ALL
TV
P52
CMP - STAR/LUNAR HORIZON SIGHTING
CMP - STAR/EARTH HORIZON SIGHTINGS
CDR SLEEP PERIOD (7 HOURS)
132:00

132:00
PTC
CMP & LMP EAT PERIOD
CDR EAT PERIOD
CDR SLEEP PERIOD (7 HOURS)
P52
PTC
138:00
CMP & LMP SLEEP PERIOD (7 HOURS)
P52
PTC
CDR EAT PERIOD
CMP & LMP EAT PERIOD
144:00
P52

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
		November 22, 1968	120:00 - 144:00	6	5-6

FLIGHT PLAN

Time		CDR	CMP	LMP	MCC-H
143:00					
					P27 UPDATE: STATE VECTOR (LM & CSM SLOTS) REFSMMAT
EI-3.5 HRS				RECORD MNVR AND ENTRY PAD	VOICE UPDATE: MNVR PAD ENTRY PAD
143:30	M S F N	MNVR TO P52 ATT			
			IMU REALIGN P52 OPTION 3 - REFSMMAT STAR ID ___.___ ___ STAR ANGLE DIFF ___.___ ___ TORQUE ANGLES: X ___ ___.___ ___ ___ Y ___ ___.___ ___ ___ Z ___ ___.___ ___ ___	ECS CK	
EI-3 HRS				EPS CK SPS CK SM/CM RCS CK C&W CK	
			CMC SELF CK DSKY COND LT TEST		
144:00					

BURN STATUS REPORT

X X □ : .	ΔTIG
X X .	BT
□ .	V_{gx}
TRIM	
X X X	R
X X X	P
X X X	Y
□ .	V_{gx}
□ .	V_{gy}
□ .	V_{gz}
□ .	ΔV_c
X X X	FUEL
X X X	OX
X X X	UNBALANCE

REMARKS:

2-114a

MCC'S

BURN CHART

	P or Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
MCC(ALL)	10°/SEC TAKEOVER	10° TAKEOVER	B/T +1 SEC	TRIM TO 0.2 fps

FLIGHT PLAN

TIME		CDR	CMP	LMP	MCC-H
144:00	MSFN	EXT ΔV P30			
EI-2.5 HRS					
144:30		SPS/RCS THRUST P40/41 MNVR TO BURN ATT	COAS & SXT STAR CK		
		GDC ALIGN TO IMU			
EI-2 HRS		$\boxed{MCC_7 \quad \Delta V\text{=NOMINALLY ZERO}}$		MCC_7 BURN STATUS REPORT	
		V66 TRANS CSM STATE VECTOR TO LM SLOT			
145:00					

Time	CDR	CMP	LMP	MCC-H
145:00		LOSS OF COMM CONTINGENCY	BIOMED Sw - LEFT	
		CISLUNAR NAVIGATION P23 1. STAR 33 ENH 00110 STAR__ __ E__H 1 SET		
	MNVR TO ENTRY ATT COAS STAR CK	SXT STAR CK		
EI-1.5 HR		IMU REALIGN P52 OPTION 3 - REFSMMAT AND GYRO DRIFT TEST STAR ID __ __.__ __ STAR ANGLE DIFF __.__ __ TORQUE ANGLES: X __ __.__ __ __ Y __ __.__ __ __ Z __ __.__ __ __		PIPA BIAS CK
145:30	GDC ALIGN TO IMU			
	FINAL STOWAGE	FINAL STOWAGE INITIATE CM RCS PREHEAT	FINAL STOWAGE	
		WASTE H_2O DUMP - OFF UR DUMP HT - OFF		
EI-1 HR	EMS CK			
146:00	FINAL GDC DRIFT CK			

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	145:00 - 146:00	7/TEC	2-116

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FLIGHT PLANNING BRANCH

FLIGHT PLAN

Time	CDR	CMP	LMP	MCC-H
146:00		TERM CM RCS PREHEAT	PYRO BAT CK	P27 UPDATE: STATE VECTOR VOICE UPDATE: ENTRY PAD & RCVY INFO
	SET DET TO EI EMS INITIALIZATION RSI ALIGN TO GDC CM RCS CK	RECORD ENTRY PAD & RCVY INFO		
EI-30 MIN			ENTRY BATS - ON	
	SEPARATION CK LIST	SEPARATION CK LIST	SEPARATION CK LIST	
146:30	MNVR TO CM/SM SEP ATT	P61 ENTRY PREP	GO FOR PYRO ARM	GO FOR PYRO ARM
EI-15 MIN	CM/SM SEP MNVR TO ENTRY ATT	P62 ENTRY ATTITUDE		
		P63 ENTRY INITIATE		
146:50	EI = 400K	P64 ENTRY POST .05G		
147:00				

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

CDR CMP LMP

147:00

EARTH LANDING PHASE

SPLASHDOWN

147:30

148:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	147:00 - 148:00	7/TEC	2-118

MSC Form 1910 (Nov 68) FLIGHT PLANNING BRANCH

FLIGHT PLAN

144:00

EI -2 HRS MCC-7 P52

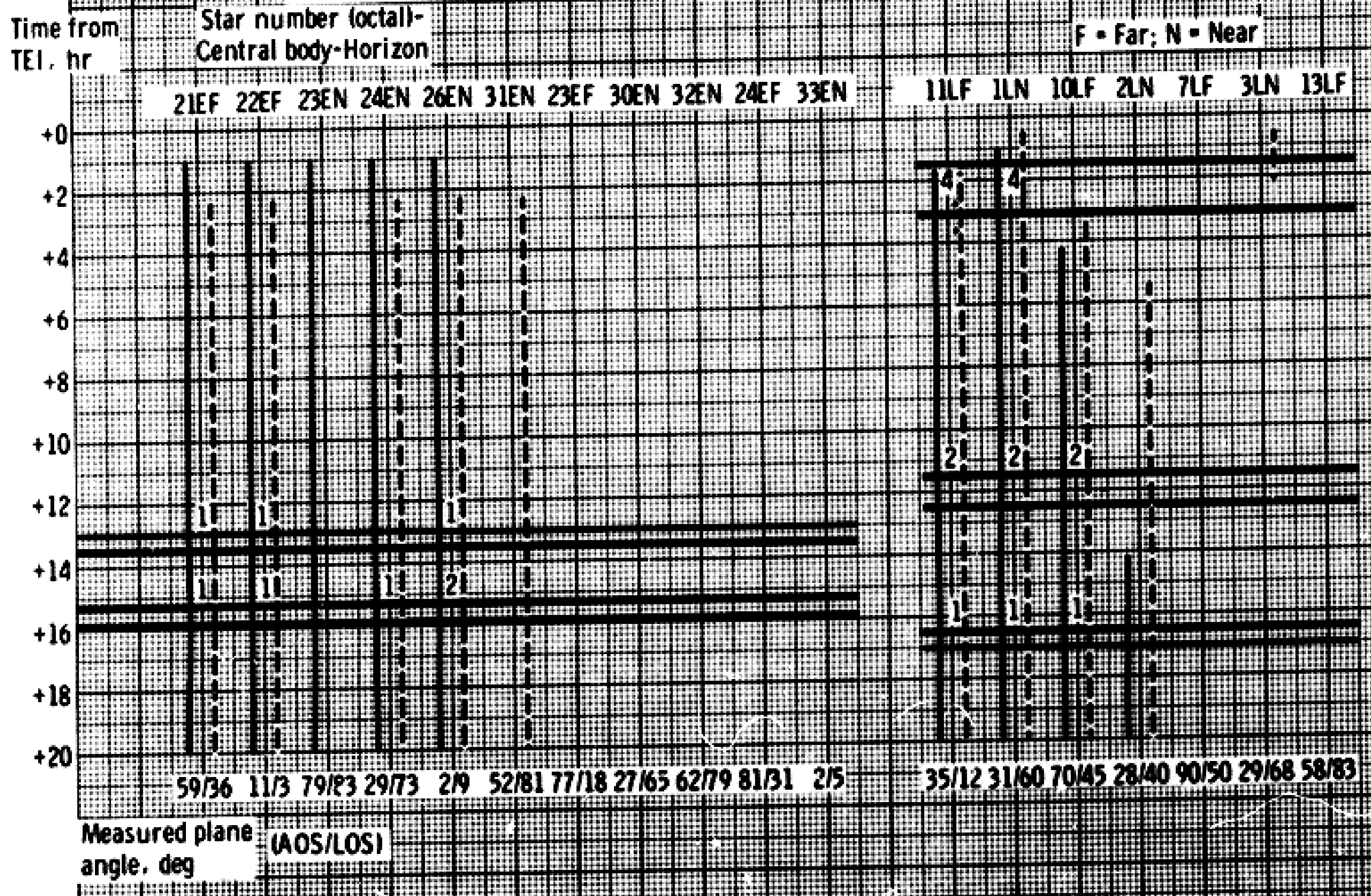
CM/SM SEP

EI = 400K

147:10 SPLASHDOWN

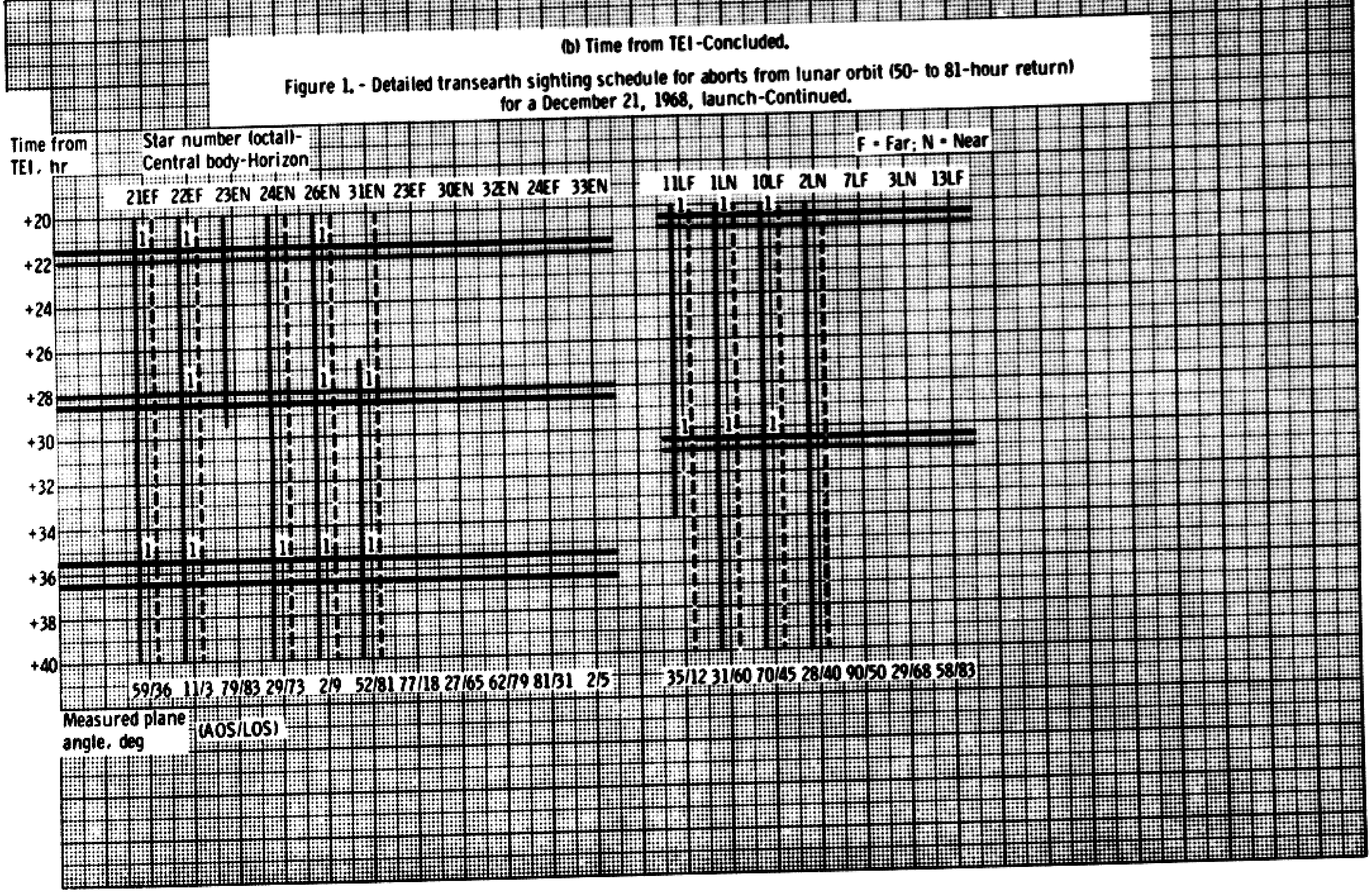
148:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL	November 22, 1968	144:00 to 146:50	6	5-7



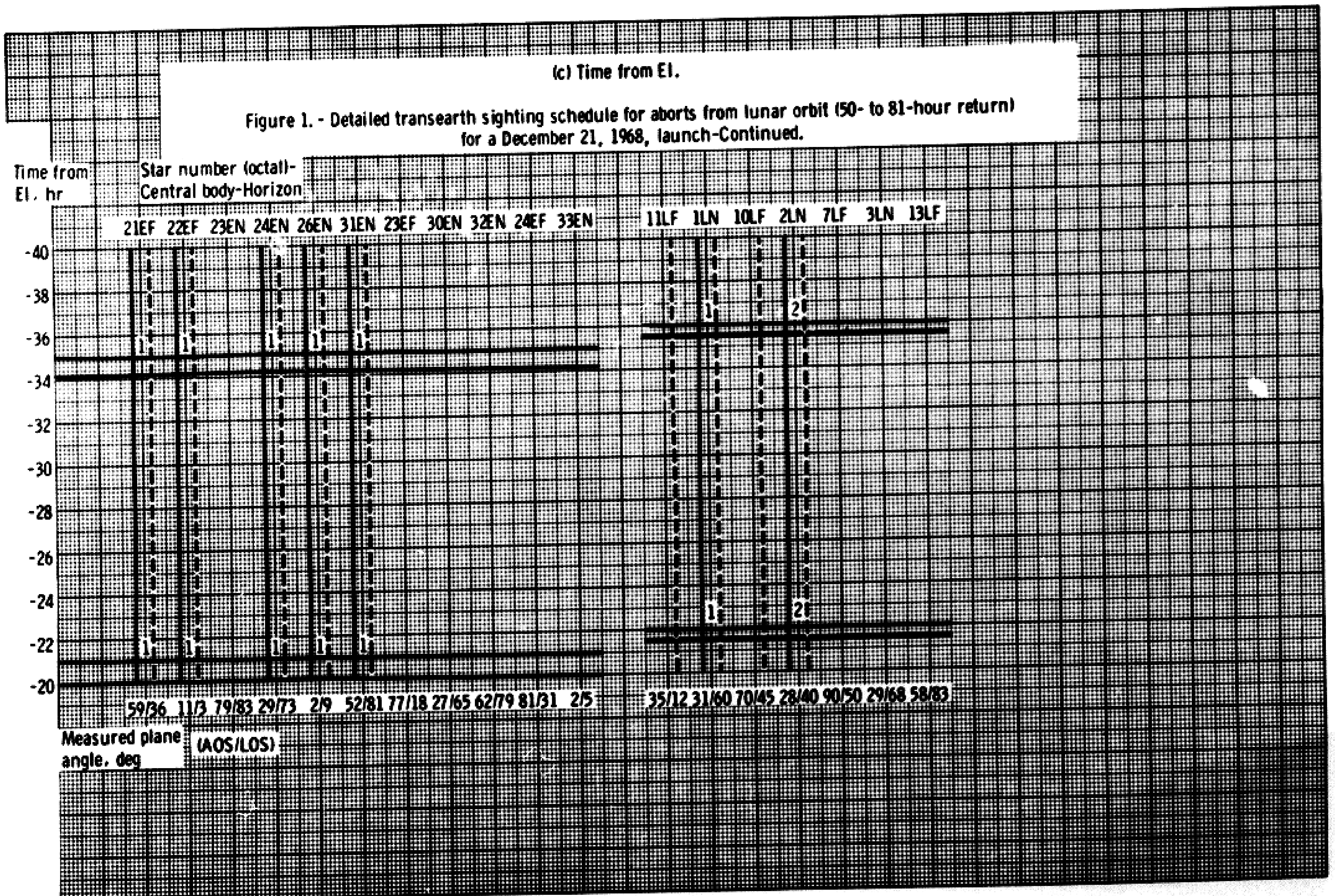
(a) Time from TEI.

Figure 1. - Detailed transearth sighting schedule for aborts from lunar orbit (50- to 81-hour return) for a December 21, 1968, launch.



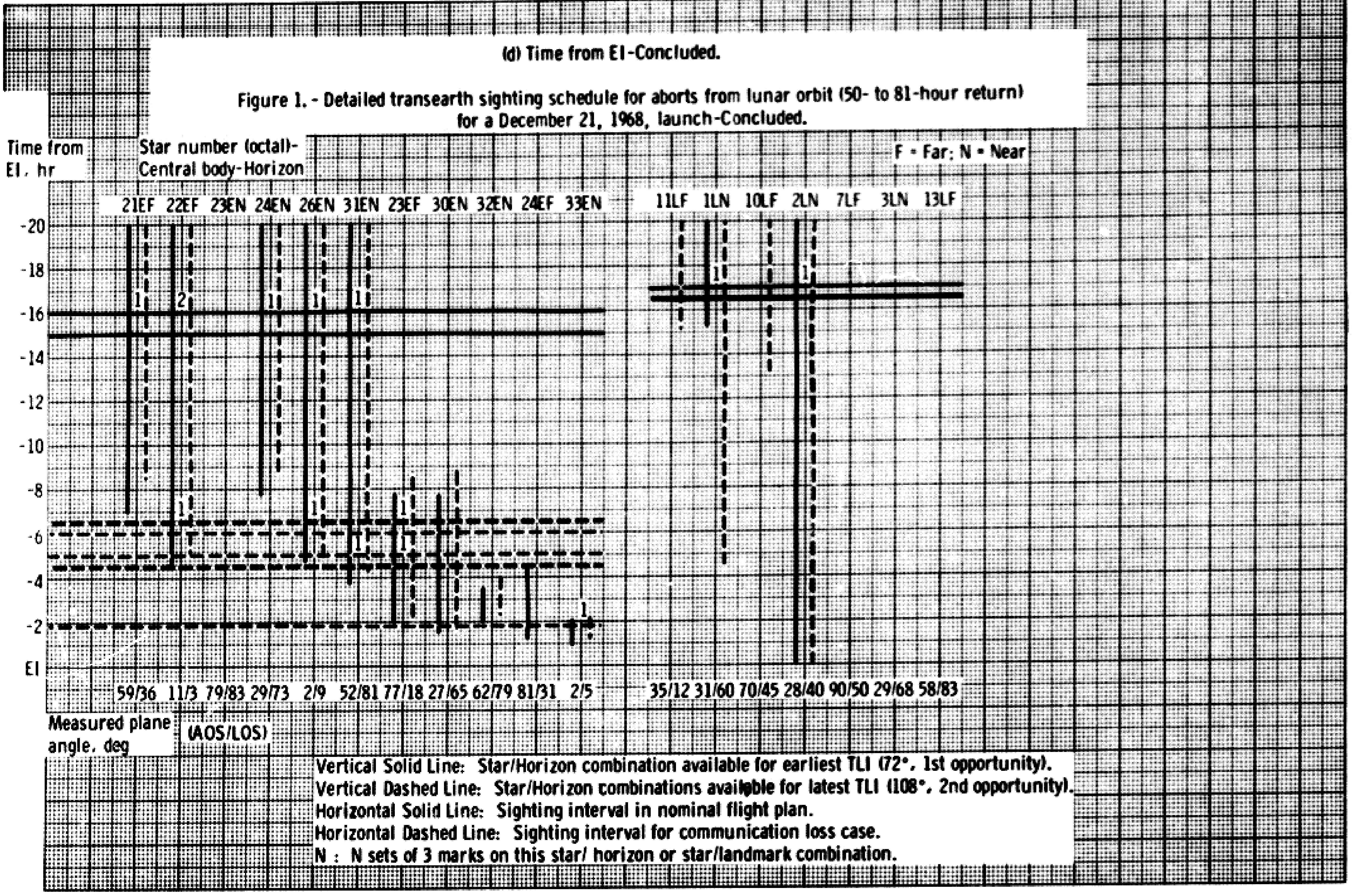
(b) Time from TEI-Concluded.

Figure 1. - Detailed transearth sighting schedule for aborts from lunar orbit (50- to 81-hour return) for a December 21, 1968, launch-Continued.



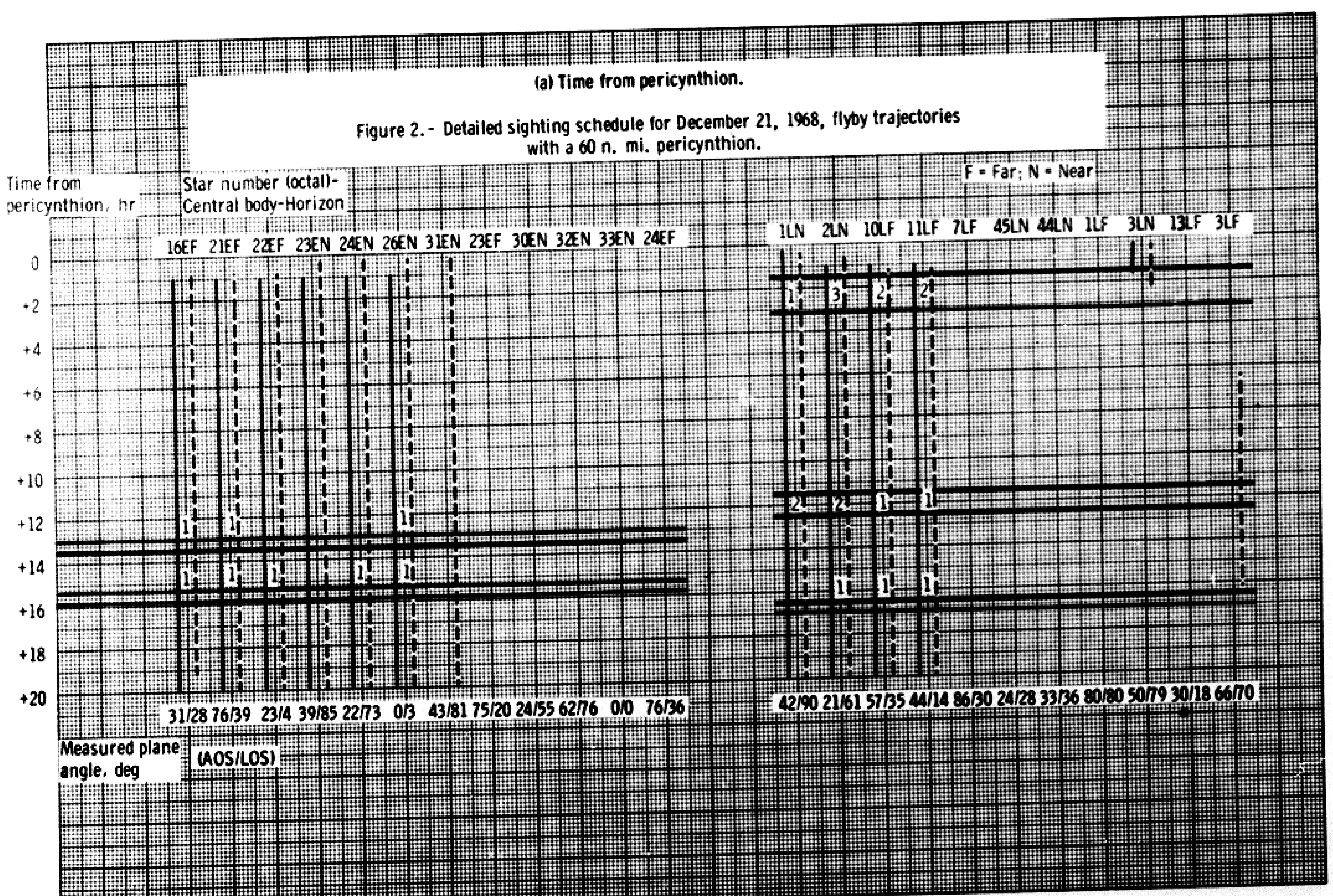
(c) Time from EI.

Figure 1. - Detailed transearth sighting schedule for aborts from lunar orbit (50- to 81-hour return) for a December 21, 1968, launch-Continued.



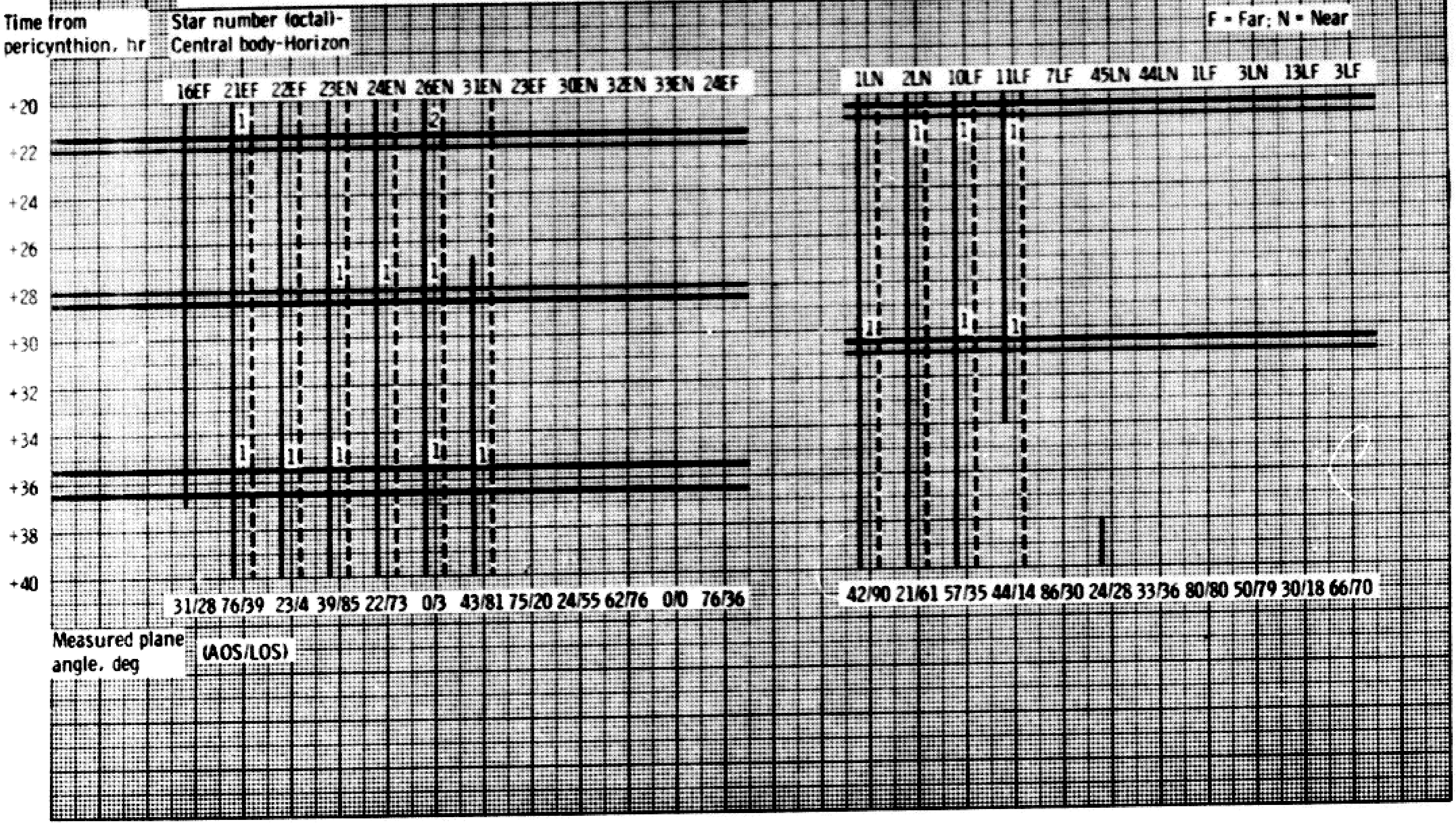
(d) Time from EI-Concluded.

Figure 1. - Detailed transearth sighting schedule for aborts from lunar orbit (50- to 81-hour return) for a December 21, 1968, launch-Concluded.



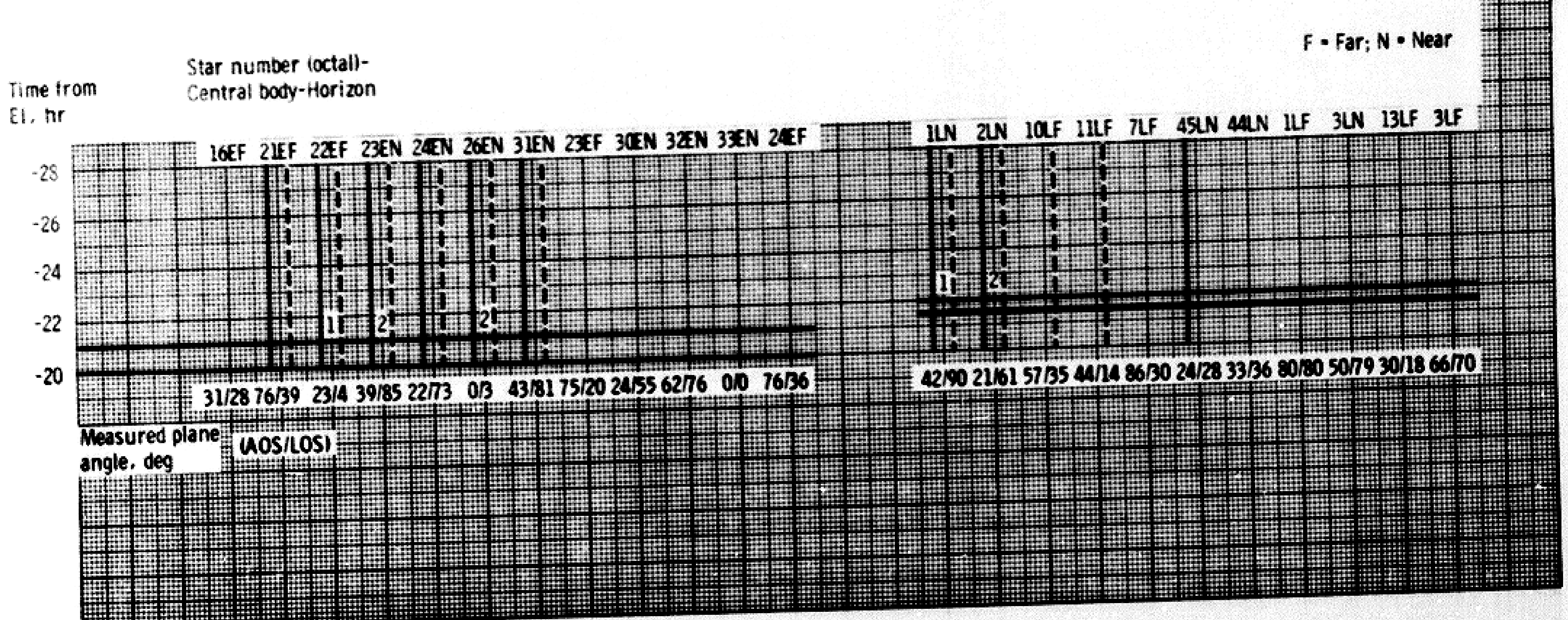
(a) Time from pericynthion.

Figure 2. - Detailed sighting schedule for December 21, 1968, flyby trajectories with a 60 n. mi. pericynthion.



(b) Time from pericynthion-Concluded.

Figure 2.- Detailed sighting schedule for December 21, 1968, flyby trajectories with a 60 n. mi. pericynthion-Continued.

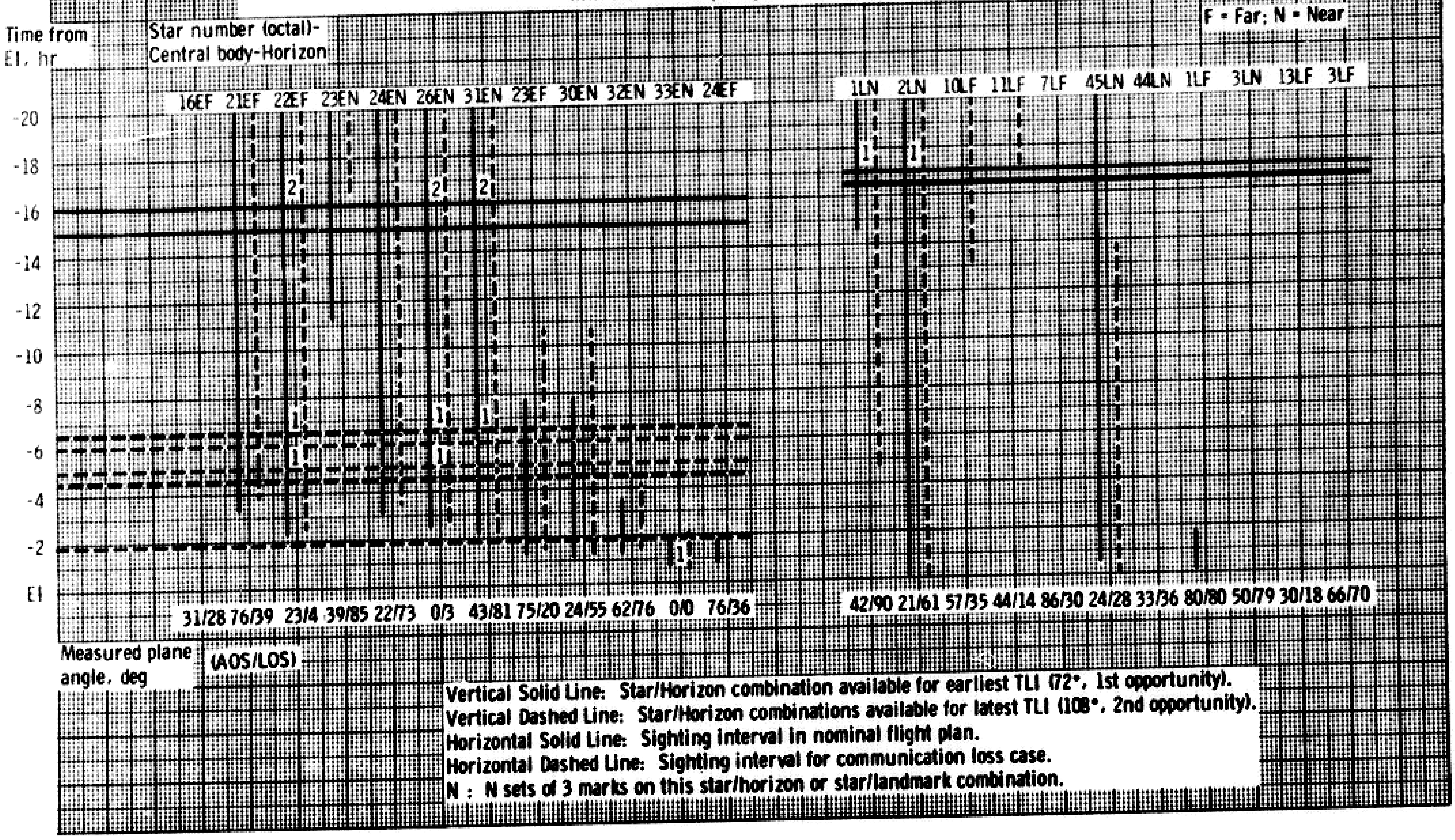


(c) Time from EI.

Figure 2.- Detailed sighting schedule for December 21, 1968, flyby trajectories with a 60 n. mi. pericynthion-Continued.

(d) Time from EI-Concluded.

Figure 2.- Detailed sighting schedule for December 21, 1968, flyby trajectories with a 60 n. mi. pericynthion-Concluded.



DEC 21 LUNAR FLYBY
TEC = 67 HOURS

FLIGHT PLAN

TIME FROM PC

00:00 PERICYNTHIAN

EAT PERIOD (ALL)

P52
CMP - 8 SETS OF STAR/LUNAR HORIZON SIGHTINGS
PTC

CDR SLEEP PERIOD (5 HOURS)

06:00

CMP & LMP EAT PERIOD

CDR EAT PERIOD

CMP & LMP SLEEP PERIOD (5 HOURS)

12:00

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AS503/103	FINAL			TEC	1

FLIGHT PLAN

12:00

EAT PERIOD (ALL)

P52

CMP - 6 SETS OF STAR/LUNAR HORIZON SIGHTINGS

CMP - 3 SETS OF STAR/EARTH HORIZON SIGHTINGS

PC+17 MCC 5

EAT PERIOD (ALL)

18:00

P52

CMP - 5 SETS OF STAR/EARTH HORIZON SIGHTINGS

CMP - 3 SETS OF STAR/LUNAR HORIZON SIGHTINGS

PTC

CDR SLEEP PERIOD (6 HOURS)

CMP & LMP EAT PERIOD

24:00

MISSION	EDITION	DATE	TIME	DAY/REV.	PAGE

FLIGHT PLAN

24:00

CDR
EAT PERIOD

CMP & LMP
SLEEP PERIOD
(6 HOURS)

30:00

EAT PERIOD
(ALL)

P52
CMP - 3 SETS OF
STAR/LUNAR
HORIZON SIGHTINGS

CMP - 3 SETS OF
STAR/EARTH
HORIZON SIGHTINGS

MCC 6

EAT PERIOD
(ALL)

CMP - 3 SETS OF
STAR/LUNAR
HORIZON SIGHTINGS

PTC

36:00

MISSION	EDITION	DATE	TIME	DAY/REV.	PAGE

FLIGHT PLAN

36:00

P52
CMP - 6 SETS OF STAR/EARTH HORIZON SIGHTINGS
PTC

CDR
SLEEP PERIOD
(6 HOURS)

CMP & LMP
EAT PERIOD

CDR
EAT PERIOD

42:00

CMP & LMP
SLEEP PERIOD
(6 HOURS)

EAT PERIOD
(ALL)

P52

48:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

FLIGHT PLAN

48:00

CMP - 3 SETS OF STAR/LUNAR HORIZON SIGHTINGS

MCC 7

EAT PERIOD (ALL)

CMP - 6 SETS OF STAR/EARTH HORIZON SIGHTINGS

CDR SLEEP PERIOD (5.5 HOURS)

54:00

CMP & LMP SLEEP PERIOD (5.5 HOURS)

60:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
AS503/103	FINAL				

FLIGHT PLAN

60:00

EAT PERIOD
(ALL)

P52

CMP - 3 SETS OF
STAR/EARTH
HORIZON SIGHTINGS

CMP - 2 SETS OF
STAR/EARTH
HORIZON SIGHTINGS

MCC 6

CMP - 1 SET OF
STAR/EARTH
HORIZON SIGHTINGS

66:00

EI

LUNAR FLYBY PHOTO PROCEDURES

Hp<180NM

1/80/NEW B&W, 1/250, F2.8
BRKT, INT, CHART FOR F STOPS

* VERTICAL STRIP PHOTOGRAPHY
  WITH EXTRA EXPOSURES AS POSSIBLE

* CHANGE TO 250 LENS AT Hp+20MIN

2/250/NEW C, 1/60, F5.6, CHART

* TERMINATOR PHOTOGRAPHY

* PHOTOGRAPH TARGETS AND GENERAL
  SCENE AS RAPIDLY AS POSSIBLE

* REPLACE MAG WITH NEW B&W

* USE R&B ON TARGET 59 AT Hp+50
  MIN: THREE PAIRS OF EXPOSURES
  AT 3, 4, AND 5 STOPS WIDER THAN SPOT

16/18/NEW CEX 1/250 F2.8 1FPS SPOT

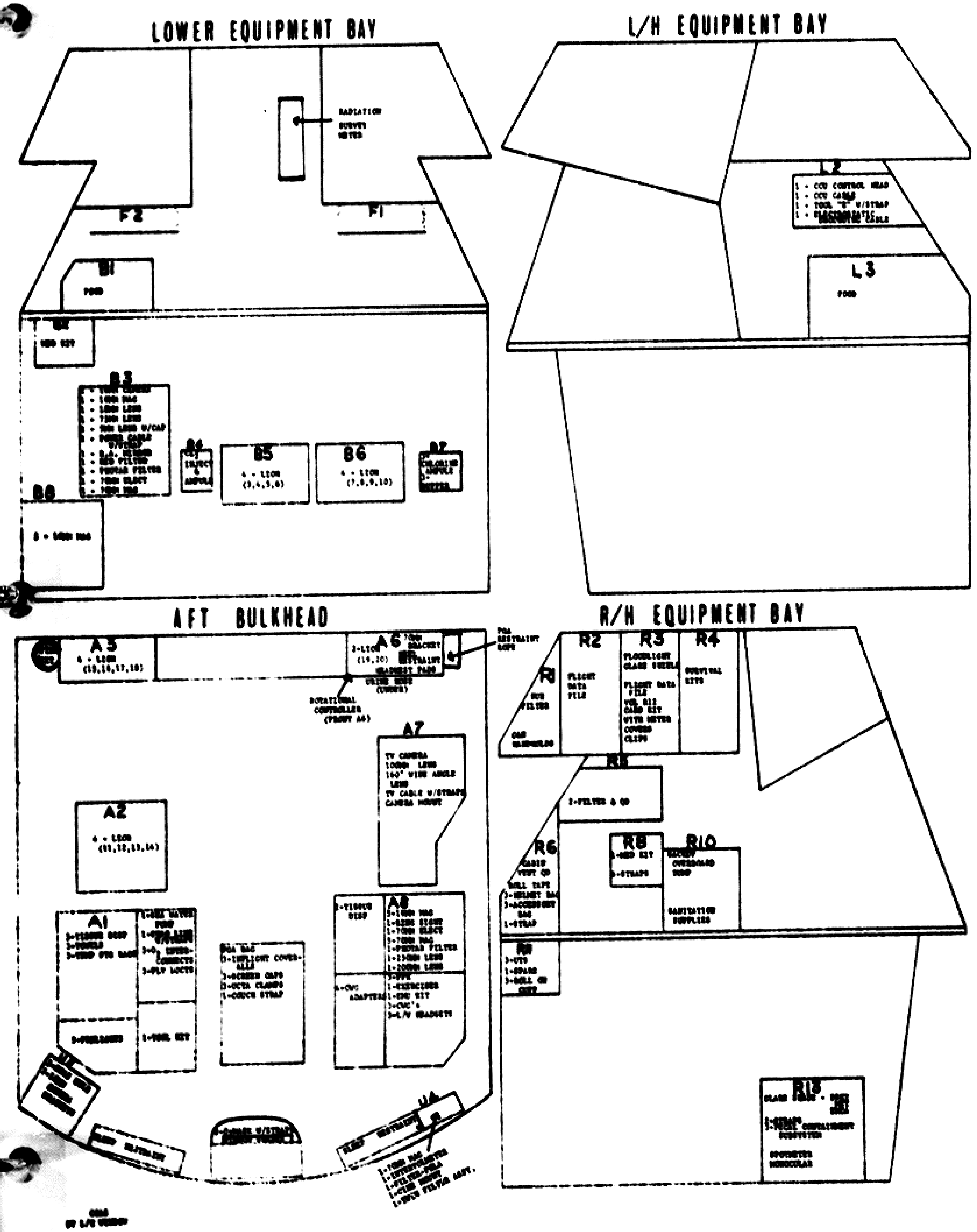
* SYSTEMATIC COVERAGE OF TERMINATORS
  AND SUNLIT SIDE

* EARTH SET AND EARTHRISE

* CHANGE TO LONGER LENS AS
  DISTANCE INCREASES

Hp>180NM

SAME AS ABOVE EXCEPT BEGIN VERTICAL
STRIP WITH 250 LENS


LOWER EQUIPMENT BAY
L/H EQUIPMENT BAY
AFT BULKHEAD
R/H EQUIPMENT BAY
F2
F1
B1
B3
B4
B5
B6
B8
L2
L3
A3
A6
A7
A2
A1
A8
R2
R3
R4
R6
R8
R10
R13

ADAPTER, CWG A-8
AMPULE, BUFFER B-4, B-7
AMPULE, CHLORINE B-4, B-7
AUTOMATIC SPOTMETER R-13

BACKUP URINE DUMP SYS R-1
BAG, TEMP STOWAGE U-1
BAGS, FECAL R-10
BRACKET, HASSELBLAD A-6
BRACKET, SEQ CAMERA U-3
BUFFER AMPULE B-4, B-7
BULB, COAS U-3

CABLE, TV A-7
CAMERA 16mm B-3
CAMERA 70mm B-3, A-8
CAMERA BRACKET U-3
CARDS, DATA R-3
CCU CABLE, SPARE L-2
CHLORINE AMPULE B-4, B-7
CINE MOUNT U-4
CLIP, DATA FILE R-3
COAS BULB U-3
CONSTANT WEAR GARMENTS A-8
CONTROL HEAD, SPARE L-2
COUPLING ASSY, 02 UMB A-1
COVERALLS, PGA BAG
CUFFS, ROLL ON R-11
CWG ADAPTER A-8

DATA ACQUISITION CAMERA B-3
DATA CARD KIT R-3
DATA FILE R-1, R-2, R-3, R-12
DUCT, PLV A-1

EARTUBE, UNIVERSAL, ICG JACKET
EMU MAINT KIT A-8
EXERCISER A-8

FECAL BAGS R-10
FECAL CONT SUBSYSTEM R-13
FILM, 16mm A-8, B-8
FILM 70mm A-8, U-4
FILTER HOLDER U-4
FILTER PHOTAR A-8, B-3
FILTER, POLA U-4
FILTER, RED B-3
FILTER, SUN R-1
FILTER, URINE R-5
FIRE EXTINGUISHER A-3
FLOODLIGHT GLARE SHIELD R-3

G&N FILTER R-1
G&N HANDHOLD R-1
GLARE SHADES, MDC R-13
GLARE SHIELD R-3

HANDHOLD, G&N R-1
HASSELBLAD CAMERA B-3, A-8
HASSELBLAD CAMERA BRACKET A-6
HEADSETS, LT. WT. A-8
HEADREST PAD A-6
HEEL RESTRAINT A-6
HELMET STOWAGE BAGS R-6
HOLDER, SLIDE & FILTER U-4
HOOK, SNAG LINE A-1

HOSE, URINE, A-6

INFLIGHT COVERALLS, PGA BAG
INFLIGHT EXERCISER A-8
INTERCONNECT, 02 HOSE A-1
INTERVALOMETER U-4

LENS, 5mm B-5
LENS, 18mm B-3
LENS, 75mm B-3
LENS, 200mm A-8
LENS, 250mm A-8
LENS, TV A-7
LIGHT BULD, COAS U-3
LIGHTWEIGHT HEADSETS A-8
LUNAR SURFACE FILM MAG A-8, U-4

MAGAZINE, 16mm A-8, B-8
MAGAZINE, 70mm A-8, U-4
MAINT KIT, EMU A-8
MEDICAL KIT B-2
MIRROR RT ANGLE B-3
MONOCULAR R-13
MOUNT, 16mm CAMERA U-3
MOUNT, ROT CONT A-6

NASAL EMOLLIENT B-2
NEEDLE CHLORINE B-4

02 HOSE CAPS, PGA BAG
OXY MASK, UNDER HATCH

PAD, HEADREST A-6
PENLIGHTS A-1
PHOTAR FILTER A-8, B-3
PLV DUCT A-1
POLA FILTER U-4
POWER CABLE B-3
PPK A-8
PUMP, SEA WATER A-1

Q.D. CABIN VENT R-6

RADIATION SURVEY METER, LEB
RECEIVER, UTS R-11
RED FILTER B-3
RESTRAINT, HEEL A-6
RETAINER STRAPS R-6
RIGHT ANGLE MIRROR B-3
RING SIGHT
ROLL ON CUFFS R-11
ROPE, SLEEP RESTRAINT A-6
ROT CONT MOUNT A-6

SCREEN CAP, 02 HOSE PGA BAG
SEAWATER PUMP A-1
SHADES, GLARE, MDC R-13
SHADES, WINDOW 02 RACK
SIGHT, RING A-8
SLEEP RESTRAINT ROPE A-6
SLIDE HOLDER U-4
SNAG LINE HOOK A-1
SPOTMETER R-13
STRAP, COUCH RESTRAINT, PGA BAG
STRAPS, RETAINER R-8
SURVIVAL KITS R-4
SYRINGE, CHLORINE B-4

TAPE R-6
TEMP STOWAGE BAG U-1
TISSUES A-1, A-8
TOOL SET A-1
TOWELS A-1
TV CABLE A-7
TV CAMERA A-7
TV CAMERA MOUNT A-7
LENS A-7

UCTA CLAMPS, PGA BAG
UNIVERSAL EARTUBE, ICG JACKET
URINE DUMP, BACKUP R-10
URINE FILTER R-5
URINE HOSE A-6
URINE TRANS SYS, R-11
UTS RECEIVER R-11

16mm Lenses B-3
16mm Mirror B-3
70mm Camera B-3, A-8
70mm Film B-3, A-8, U-4
250mm Lens A-8
200mm Lens A-8
16mm Bracket U-3
16mm Camera B-3
16mm Camera Mount U-3
16mm Film B-3, B-8, A-8
16mm Pwr Cable B-3

FLIGHT PLAN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE

FLIGHT PLANNING BRANCH

MSC FORM 1186 (SEP 67)

FLIGHT PLAN

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